



Water Monitoring and Truck Tracking Web System

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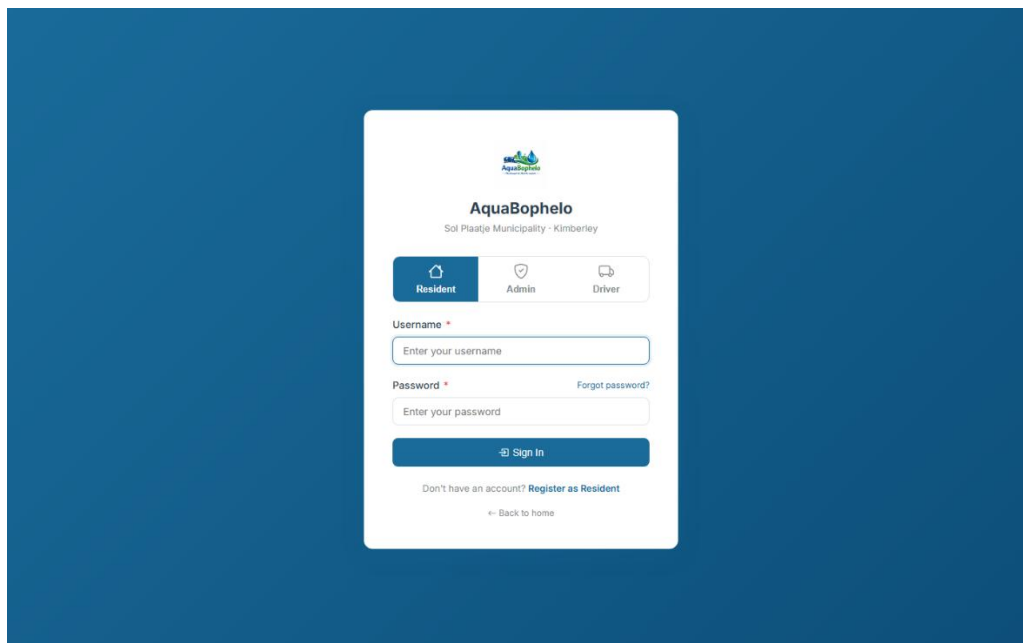
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Login / Registration / Authentication Screens

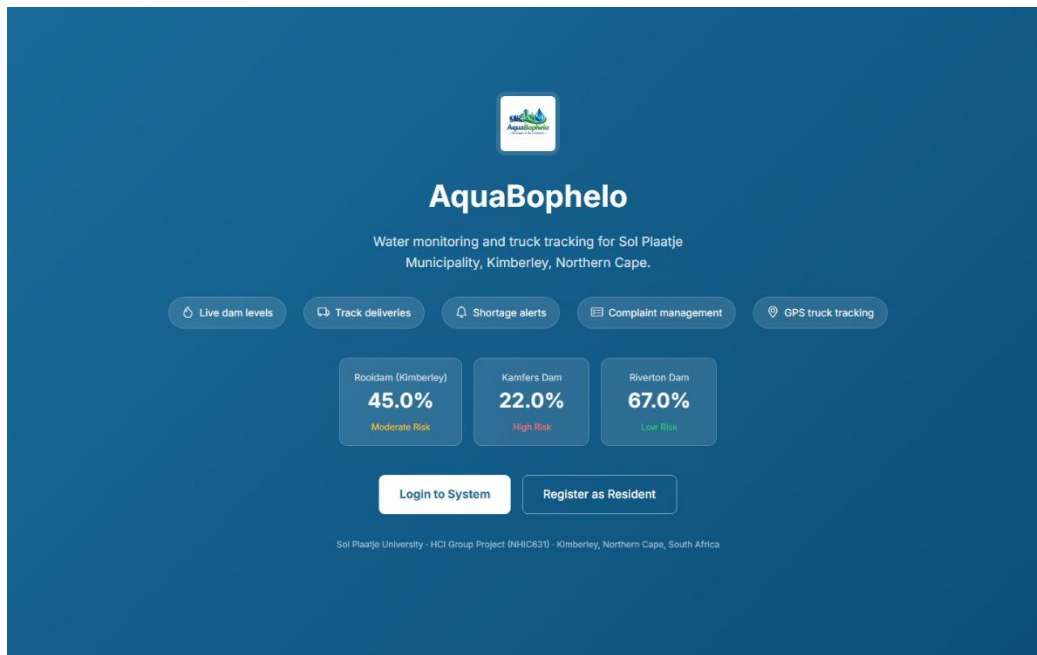
1.splash Screen

The image shows a login screen for AquaBophelo, Sol Plaatje Municipality - Kimberley. The screen has a dark blue background. In the center is a white card with the AquaBophelo logo at the top. Below the logo, the text 'AquaBophelo' and 'Sol Plaatje Municipality - Kimberley' are displayed. There are three tabs: 'Resident' (selected), 'Admin', and 'Driver'. Below the tabs are two input fields: 'Username' and 'Password'. The 'Username' field has a placeholder 'Enter your username'. The 'Password' field has a placeholder 'Enter your password' and a 'Forgot password?' link. Below the password field is a blue 'Sign In' button. At the bottom of the card, there is a link 'Don't have an account? Register as Resident' and a link 'Back to home'.

- **Description:** A simple login interface containing role selection tabs, username and password fields, a sign-in button, and account recovery options.

- **Design Choice:** Clear User Navigation. The role tabs for Resident, Admin, and Driver help users quickly identify the correct login section and reduce navigation confusion.

2. Welcome Screen



- **Description:** A centralized landing page displaying system features, live dam levels, risk indicators, and quick access buttons for login and registration.
- **Design Choice:** Information Visibility and Accessibility. Important system functions and water monitoring information are displayed immediately to help users understand the system at a glance.

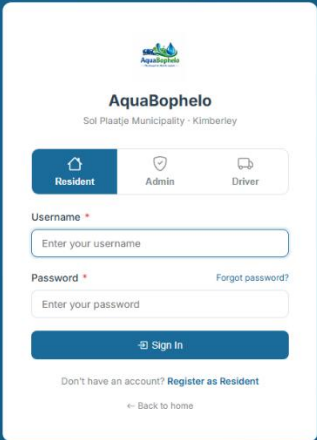
3. Registration Screen

The registration form is titled "Create resident account" with the subtitle "Register to access water services and delivery tracking". It contains the following fields: "First Name" and "Last Name" (each with a placeholder "First name" or "Last name"), "Username *" (placeholder "Choose a username"), "Email Address *" (placeholder "your@email.com"), "Phone Number" (placeholder "e.g. 076 123 4567"), and "Your Area *" (a dropdown menu with "-- Select your area --"). At the bottom are "Password *" and "Confirm Password *" fields (placeholders "Create password" and "Confirm password" respectively), a "Create Account" button, and a link "Already have an account? Sign In".

- **Description:** A structured account registration form allowing residents to enter personal details, select their area, and create login credentials.

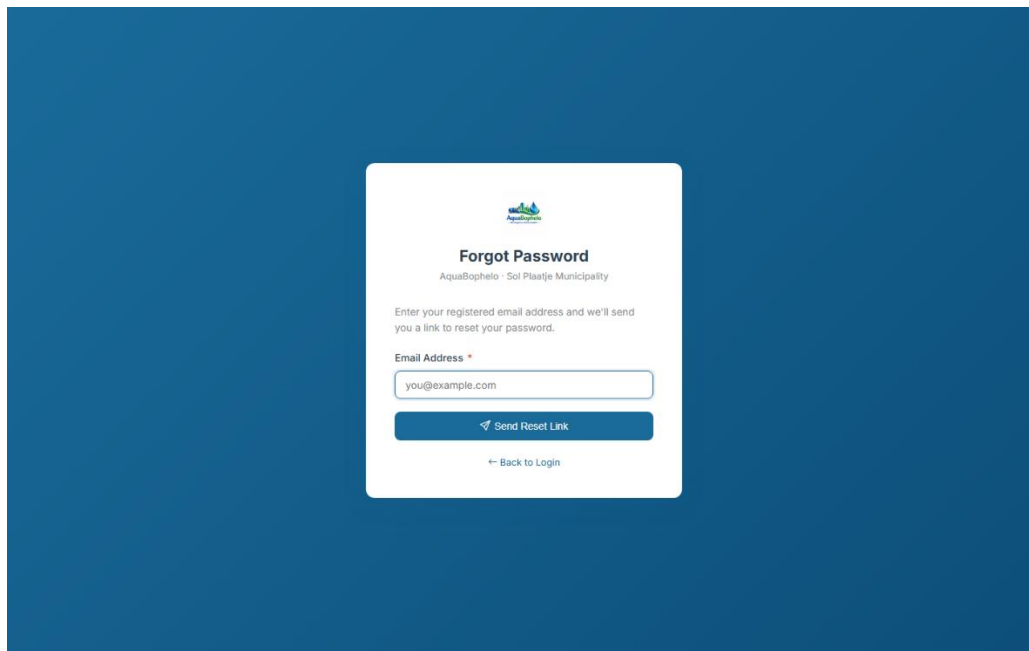
- **Design Choice:** Structured Data Collection. Related fields are grouped together to make the registration process simple, organized, and easy to complete.

4. Resident/Admin/Driver Login Screen

The image shows a login screen for AquaBophelo, which is part of the Sol Plaatje Municipality in Kimberley. The interface is centered on a white card against a dark blue background. At the top of the card is the AquaBophelo logo and the text 'AquaBophelo Sol Plaatje Municipality · Kimberley'. Below this are three tabs for role selection: 'Resident' (selected), 'Admin', and 'Driver'. The 'Resident' tab is highlighted with a blue background and a house icon. The 'Admin' tab has a shield icon, and the 'Driver' tab has a car icon. Under the tabs, there are two input fields: 'Username *' with the placeholder 'Enter your username' and 'Password *' with the placeholder 'Enter your password'. To the right of the password field is a link for 'Forgot password?'. Below the input fields is a blue 'Sign In' button with a right-pointing arrow. At the bottom of the card, there is a link 'Don't have an account? Register as Resident' and a link '← Back to home'.

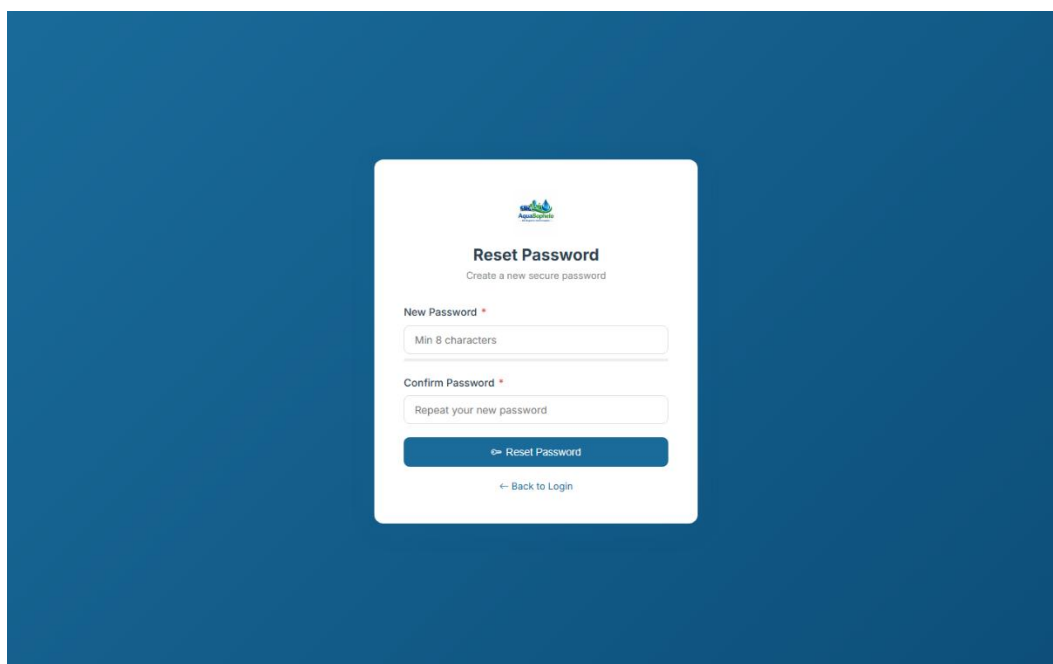
- **Description:** A secure login interface containing role selection tabs, username and password fields, a sign-in button, and links for password recovery and registration.
- **Design Choice:** Clear Role Identification and Simplicity. The tab-based role selection helps users access the correct portal quickly while the clean layout keeps the login process simple and focused.

5. Forgot Password Screen



- **Description:** A password assistance screen where users can enter their registered email address to receive a password reset link.
- **Design Choice:** Minimalist Recovery Process. The screen contains only the essential input field and action button to reduce complexity and help users recover accounts quickly.

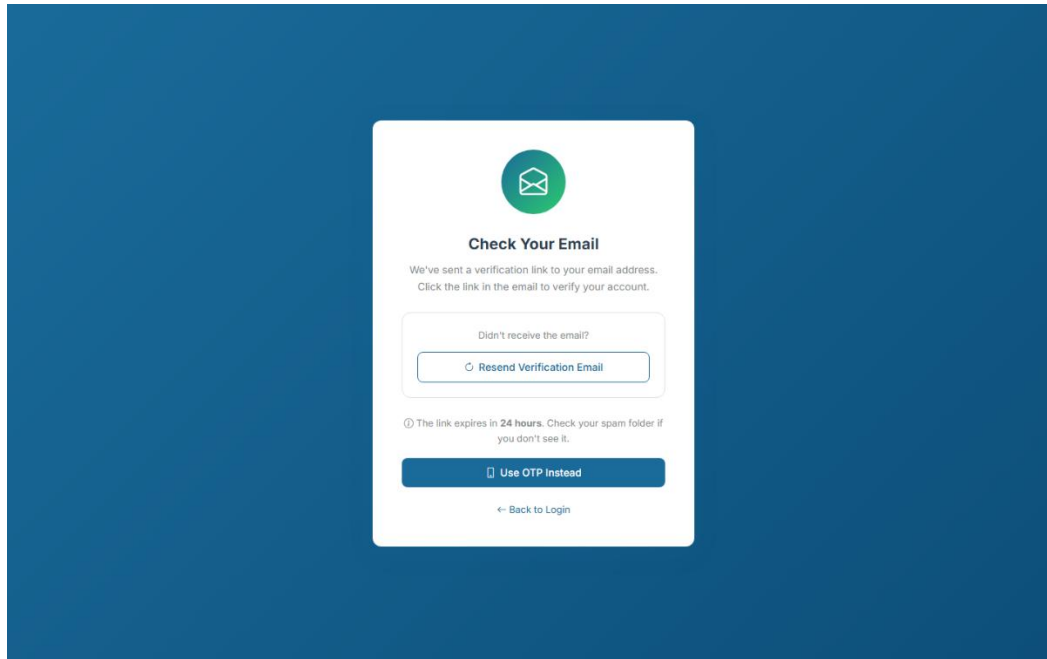
6. Reset Password Screen



- **Description:** A password recovery screen allowing users to create and confirm a new password before regaining account access.

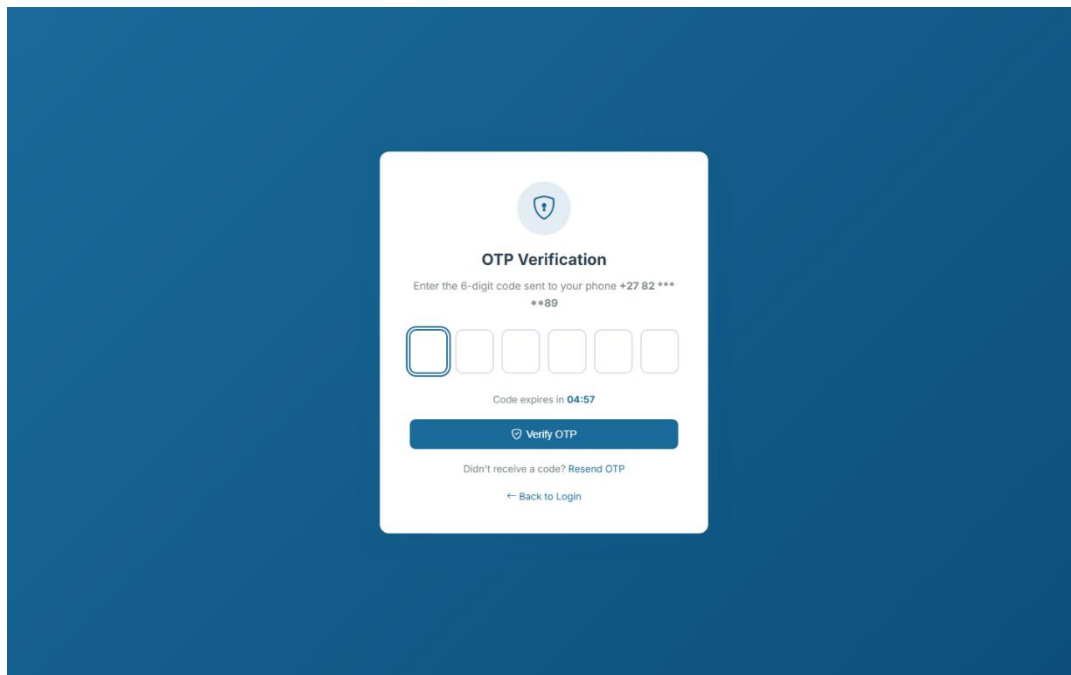
- **Design Choice:** Guided Password Recovery. The step-by-step layout with clearly labelled fields helps users complete the password reset process with minimal confusion.

7.Email Verification Screen



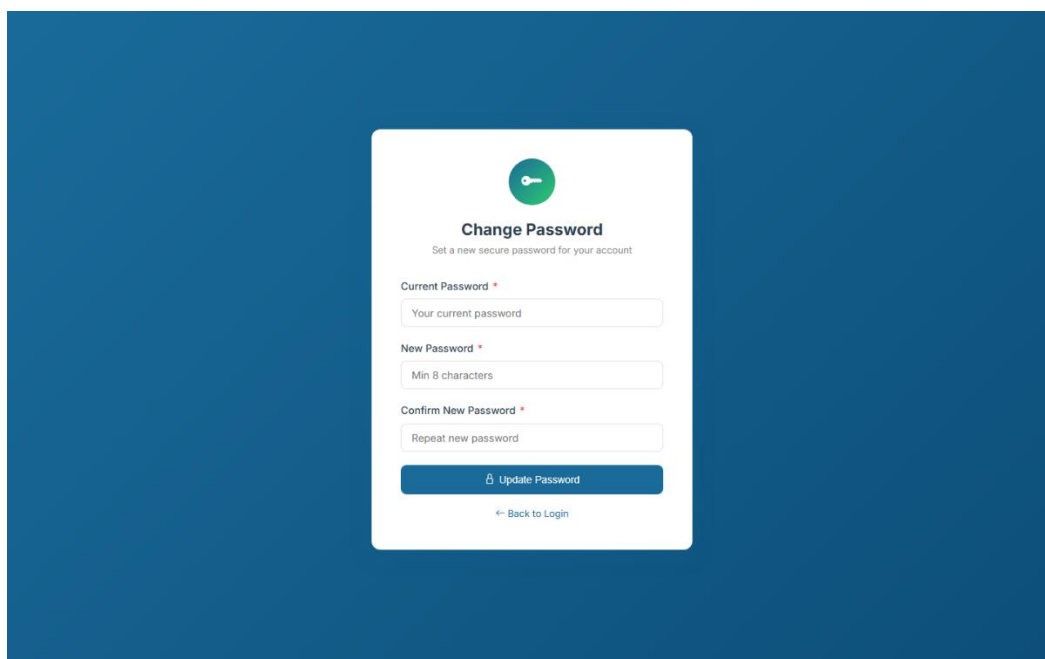
- **Description:** A confirmation interface informing users that a verification link has been sent to their email address, with options to resend the email or use OTP verification instead.
- **Design Choice:** User Guidance and Accessibility. The clean card layout and clear action buttons guide users through account verification while providing alternative verification methods for convenience.

8.OTP Verification Screen



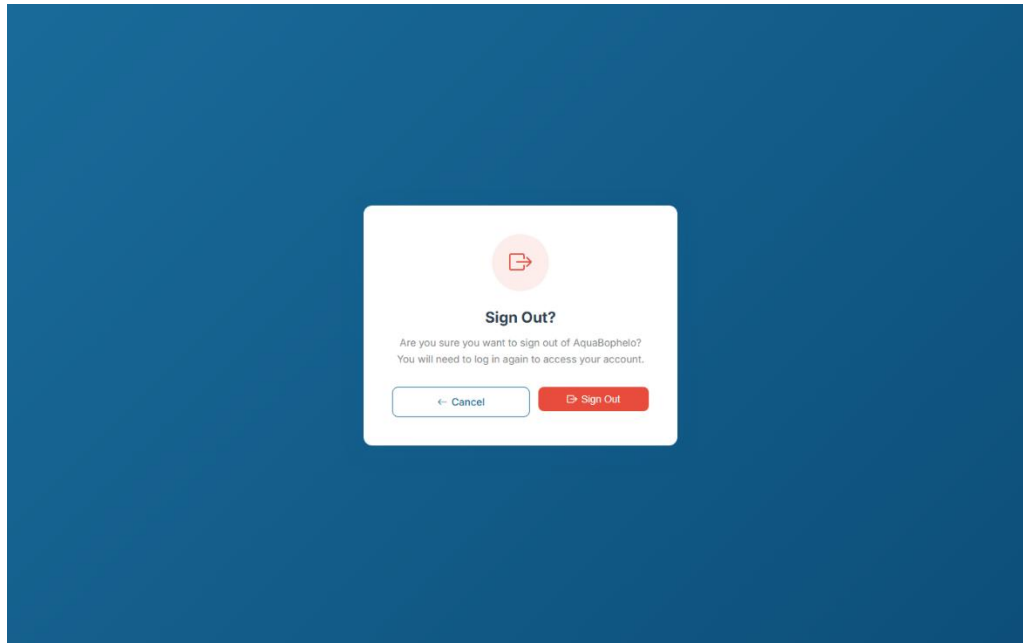
- Description: A secure OTP verification interface where users enter a six-digit code sent to their phone number, including a countdown timer and resend option.
- Design Choice: Security and Simplicity. The segmented OTP input fields improve readability and accuracy, while the timer and resend feature enhance usability during the authentication process

9.Change Password Screen



- Description: A password management interface allowing users to enter their current password and create a new secure password with confirmation fields.
- Design Choice: Security-Focused Form Design. Clearly labeled input fields and a structured layout help users securely update their passwords while minimizing input errors.

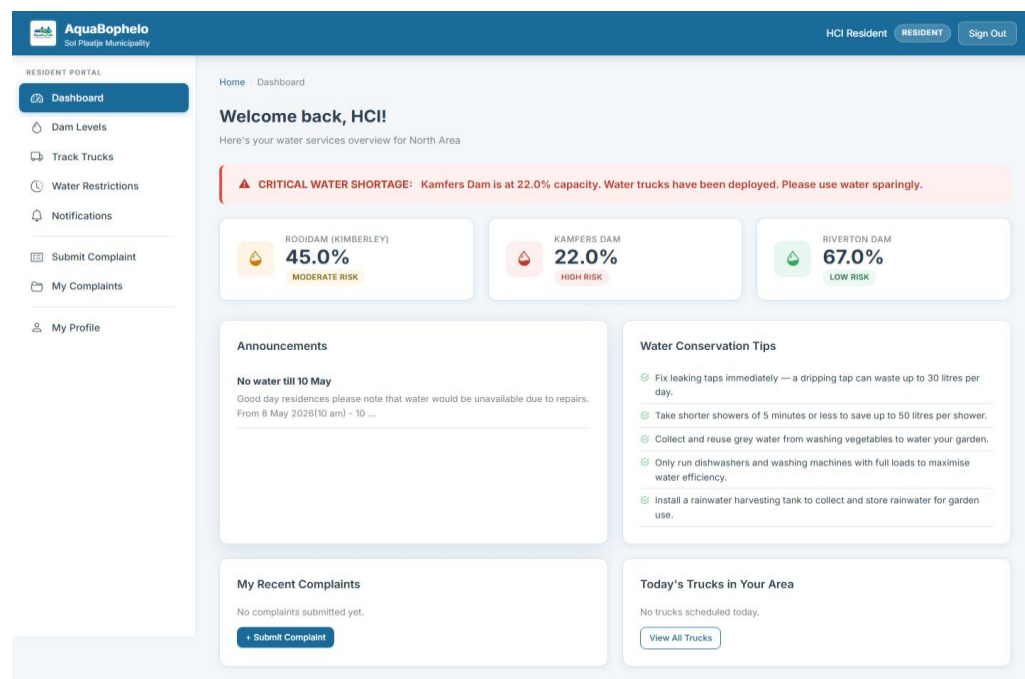
10. Logout Confirmation Screen



- Description: A confirmation dialog asking users to confirm whether they want to sign out of the system, with cancel and sign-out options.
- Design Choice: Prevention of Accidental Actions. The confirmation modal ensures users do not unintentionally log out while the contrasting button colors clearly distinguish primary and secondary actions.

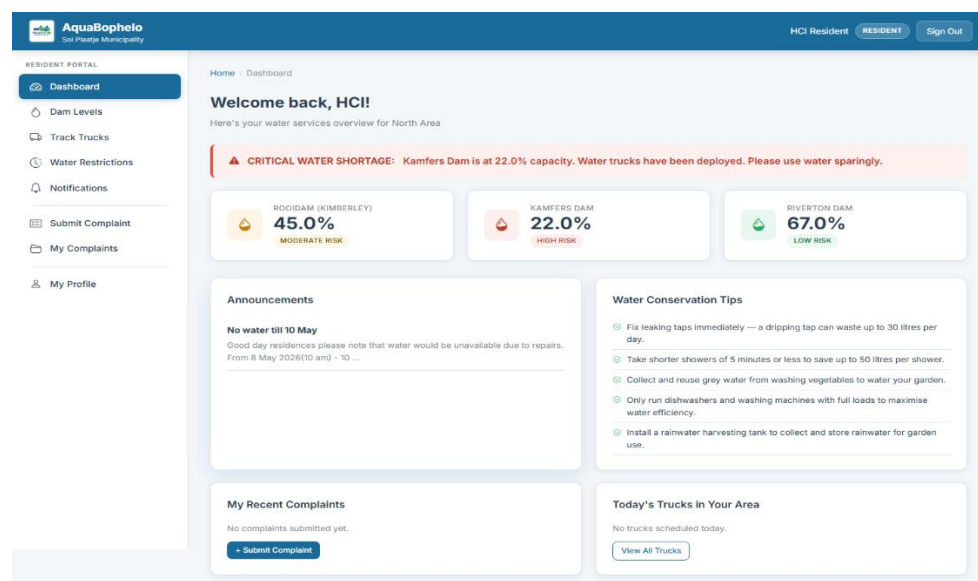
Home, Dashboard Interfaces

Resident Home Page



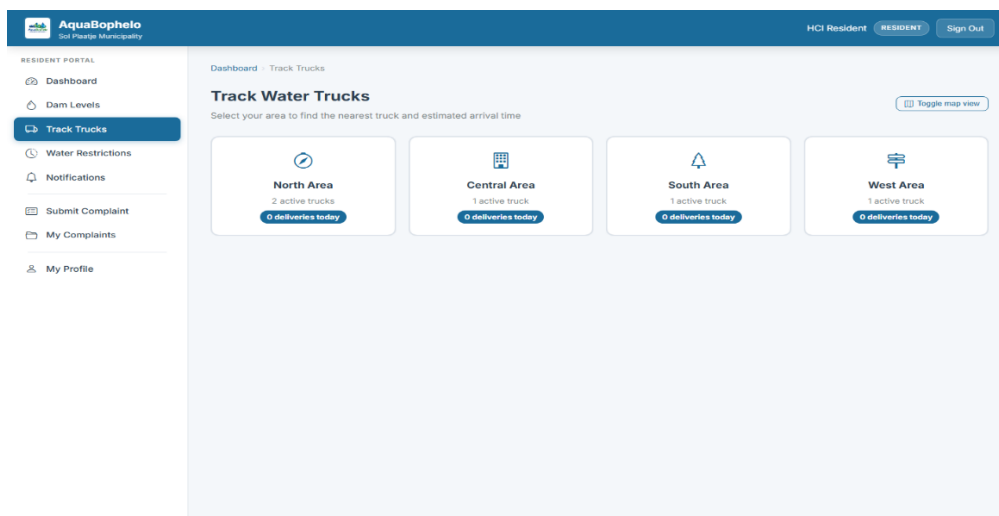
- **Description:** The landing page after login, featuring a welcoming hero section and high-level navigation to the system's core features.
- **Design Choice: User Orientation.** This screen acts as a "hub." The choice to use a clean, professional aesthetic builds trust with the resident.

Resident Dashboard



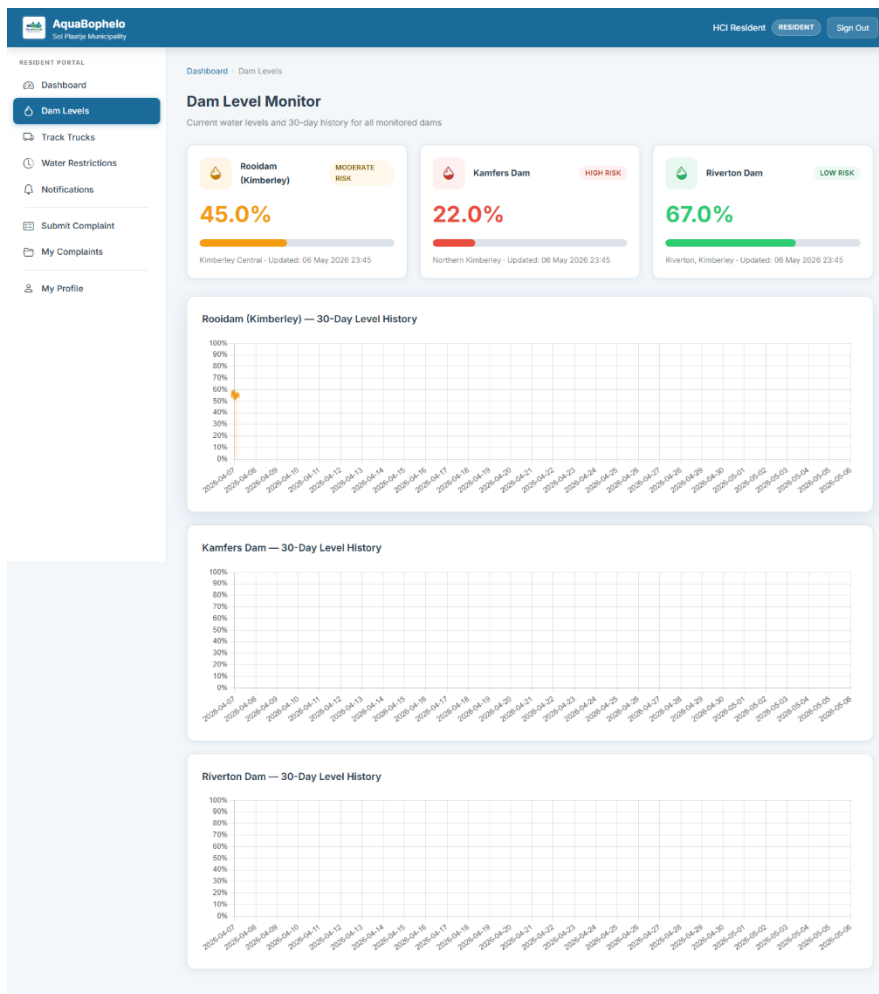
- **Description:** A centralised view showing summarised status cards for water levels, truck proximity and active alerts.
- **Design Choice: Efficiency and Scannability.** This screen avoids map-clutter. It uses a **Card-based layout** to allow users to grasp the most important information in under five seconds.

4. Water Truck Tracking Screen



- **Description:** A real-time tracking interface showing the live location of delivery trucks and estimated arrival times.
- **Design Choice: Transparency and Reduced Anxiety.** Inspired by Uber and DHL, this screen provides "Visibility of System Status." Showing the truck moving (even in simulation) reducing the user's stress by providing proof that help is on the way.

5. Dam Water Levels Screen



- **Description:** Visual representations showing current dam percentages and historical trends.
- **Design Choice: Theoretical Alignment (Predictive Awareness).** By showing *trends* and not just a single number, the design helps users perform "mental modelling" to understand if a shortage is likely in the coming days, not just on the current day.

6. Water Restrictions Screen

Dashboard · Water Restrictions

Water restriction schedule

Upcoming planned water supply interruptions by area

All Areas North Central South West

Area	Date	From	To	Reason
SOUTH	13 May 2026	10 a.m.	6 p.m.	Pressure balancing and system upgrade for the Roocepan area.

- **Description:** A clear schedule listing specific times and areas affected by water shedding or restrictions.
- **Design Choice: Clarity and Constraint.** The design uses a structured table or list format. This follows the principle of mapping, where information is organized logically by time and location to ensure there is no ambiguity for the resident.

7. Notifications & Alerts Screen

Dashboard · Notifications

Notification Centre

UNREAD: 1 Mark All Read

System Alerts

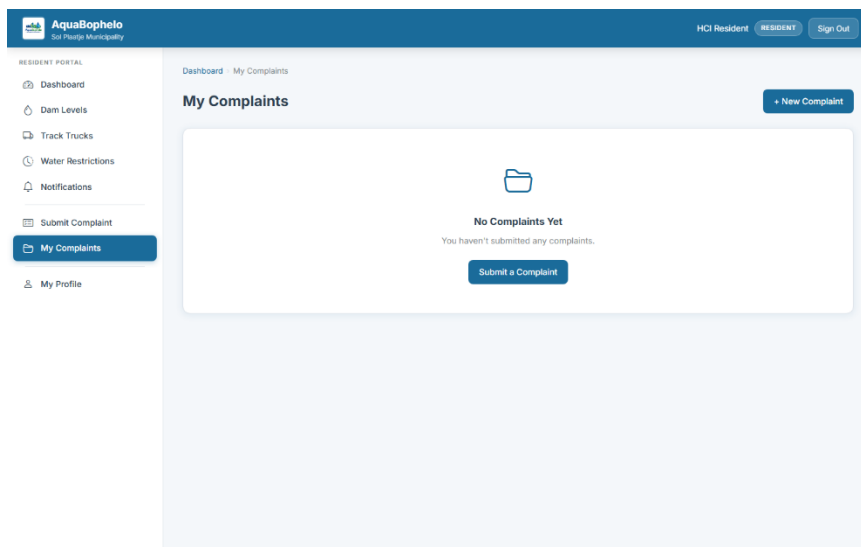
Good day residences please note that water would be unavailable due to repairs. From 8 May 2026(10 am) - 10 May 2026 (8am)
07 May 2026 09:36

Announcements

No water till 10 May
Good day residences please note that water would be unavailable due to repairs. From 8 May 2026(10 am) - 10 May 2026 (8am)
07 May 2026

- **Description:** A chronological feed of SMS and email-style alerts regarding emergencies or system updates.
- **Design Choice: Recognition over Recall.** Users don't have to remember when they received a warning; they can simply check this feed. High-contrast colours or icons (like red exclamation marks) are likely used to distinguish urgent alerts from general news.

8. Complaint Dashboard



- **Description:** An interface where residents can submit new complaints and track the "Status" (e.g., Pending, Resolved) of previous ones.
- **Design Choice: User Control and Feedback.** Providing a status tracker is a critical design choice that ensures the system provides "Feedback" to the user, letting them know the municipality is actually working on their issue.

9. Delivery Schedule Screen



AquaBophelo

Let's Change Municipality

H2I Resident

RESIDENT

Sign Out

RESIDENT PORTAL

Dashboard

Dam Levels

Track Trucks

Delivery Schedule

Water Restrictions

Notifications

Submit Complaint

My Complaints

Conservation Tips

Give Feedback

My Profile

Home

Delivery Schedule

Upcoming Deliveries

Water truck delivery schedule for your area over the next 3 days

Filter by area:

All Areas

North

Central

South

West

Date	Area	Location	Departs	ETA	Truck	Driver	Litres	Status
May 12, 2026	NORTH	Galeshewe Hall	07:00	08:30	TRE-001	B. Mokoena	5000L	ON THE WAY
May 12, 2026	CENTRAL	CSD Depot	09:00	10:00	TRE-002	L. Petersen	8000L	SCHEDULED
May 13, 2026	SOUTH	Roodenari Community	08:30	09:00	TRE-003	P. Sihole	6000L	SCHEDULED
May 13, 2026	WEST	Floors Open Area	10:00	11:30	TRE-004	T. Nkosi	4500L	SCHEDULED
May 14, 2026	NORTH	Platfontein	07:30	09:00	TRE-001	B. Mokoena	5000L	SCHEDULED

- **Description:** A calendar or list view showing when water trucks are scheduled to visit specific neighbourhoods.
- **Design Choice: Planning and Efficiency.** This screen provides early warning. By making the schedule public and easy to find, the need for residents to constantly monitor the live map is reduced.

10. Water Conservation Tips Screen

 AquaBophelo <i>Let's Change Municipality</i> H2I Resident RESIDENT Sign Out									
RESIDENT PORTAL Dashboard Dam Levels Track Trucks Delivery Schedule Water Restrictions Notifications Submit Complaint My Complaints Conservation Tips Give Feedback My Profile Home Conservation Tips Water Conservation Tips Simple steps to save water and help the community through the shortage Every Drop Counts The Northern Cape is in a water crisis. Following these tips can reduce household usage by up to 40%. All Indoor Kitchen Outdoor Recycling INDOOR Fix leaking taps A dripping tap can waste up to 15 litres of water per day. Fix leaks immediately. Saves up to 15L/day INDOOR Shorter showers Reduce shower time to 5 minutes. A standard shower uses 10 litres per minute. Saves up to 50L/shower INDOOR Full loads only Only run the washing machine or dishwasher with a full load to maximize efficiency. Saves up to 400L/load KITCHEN Boil only what you need Only boil the amount of water you need in the kettle. Saves up to 2L/day OUTDOOR Water at dawn Water your garden early morning to reduce evaporation. Avoid midday watering. Saves up to 30L/day OUTDOOR Harvest rainwater Use a rain barrel to collect roof water for garden use. Saves up to 200L/rain event OUTDOOR Wash car with a bucket Using a hosepipe to wash a car can use up to 300 litres. Use a bucket instead. Saves up to 250L/week RECYCLING Reuse grey water Reuse washing-up water or bath water for flush toilets or water plants. Saves up to 50L/day									

- **Description:** A library of instructional content and "how-to" guides for saving water.
- **Design Choice: Cognitive Load Management.** This information is kept in a separate, standalone screen so that it doesn't clutter the "Action-Oriented"

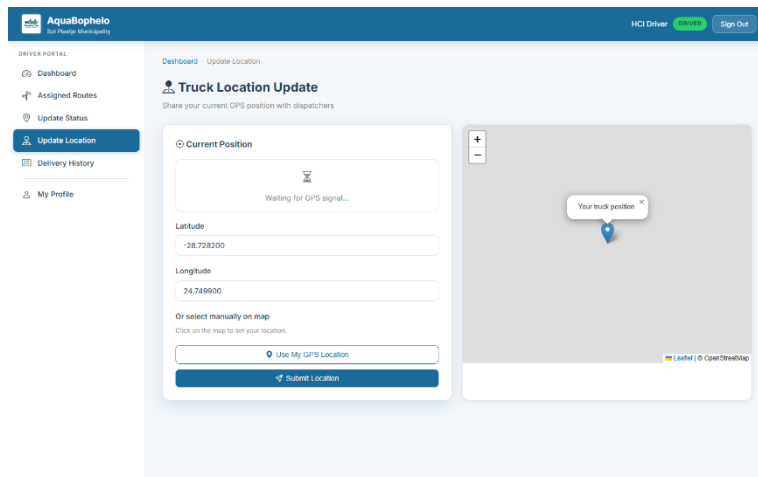
screens. It uses simple language and likely icons to ensure it is accessible to all literacy levels in the Northern Cape.

11. Resident Profile Screen

The screenshot displays the 'My Profile' screen within the AquaBophelo Resident Portal. The page features a sidebar with navigation links and a main content area. The profile card shows the user's name as 'HCI Resident' from the 'North Area'. It includes input fields for 'First Name' (filled with 'HCI'), 'Last Name' (filled with 'Resident'), 'Email Address' (filled with 'hci_resident@example.com'), and 'Phone Number' (filled with '0123456789'). A dropdown menu for 'Your Area' is set to 'North Area'. A 'Save Changes' button is located below the profile card. Below the profile card is a 'Change Password' section with three input fields: 'Old password', 'New password', and 'New password confirmation'. An 'Update Password' button is at the bottom of this section. The top right of the page indicates the user is logged in as 'HCI Resident' and provides a 'Sign Out' button.

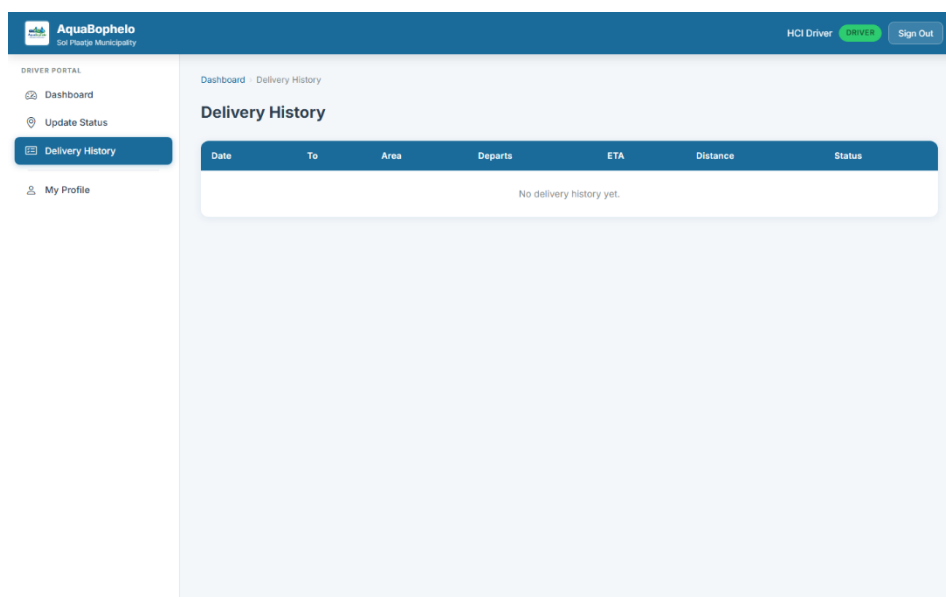
- **Description:** A personal settings hub containing the user's name, contact details (email and phone) and designated residential area, along with security options like "Change Password" and "Logout."
- **Design Choice: Data Integrity & Simplicity.** By isolating profile management from live tracking data, the design minimises cognitive load. Highlighting the "Area/Location" field ensures that residents can easily verify they are correctly categorised for targeted alerts, reducing the risk of missing critical localised information.

1. Truck Location Update Screen



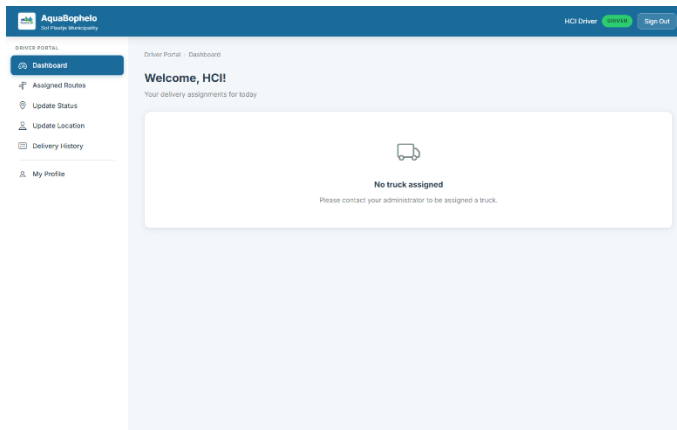
- **Description:** A live tracking interface featuring GPS coordinate fields, an interactive map, and location submission controls for updating truck positions in real time.
- **Design Choice: Visibility and User Control.** The split-screen layout allows drivers to view and update their location simultaneously, reducing confusion and improving tracking accuracy.

2.Delivery History Screen



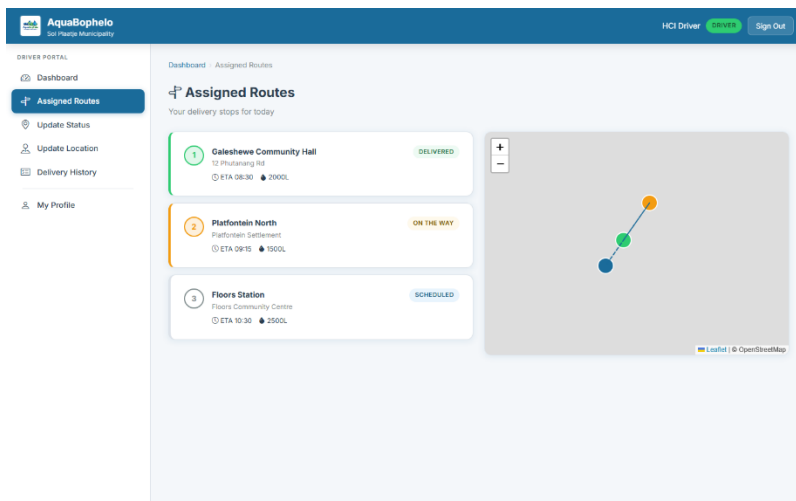
- **Description:** A structured records screen displaying delivery information such as delivery date, destination area, ETA, travel distance, and delivery status.
- **Design Choice: Efficiency and Scannability.** The table-based layout organizes delivery records into clear columns, allowing users to quickly scan and monitor delivery activities.

3.Driver Dashboard Screen



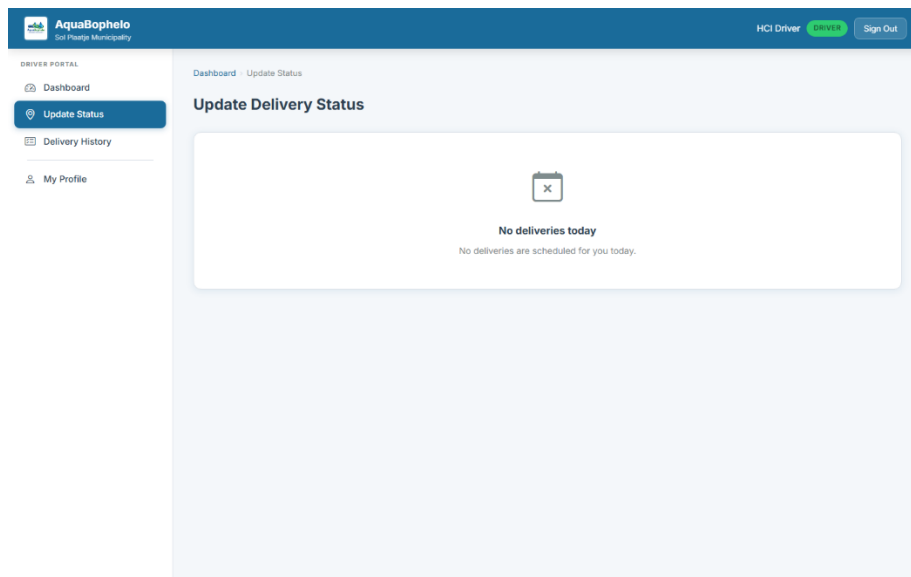
- **Description:** A dashboard screen displaying the driver portal navigation menu and delivery assignment status, including a message indicating when no truck has been assigned.
- **Design Choice: Minimal and Informative Layout.** The spacious dashboard design reduces clutter while clearly communicating assignment status and available navigation options.

4.Assigned Routes Screen



- **Description:** A route management interface displaying assigned delivery stops, estimated arrival times, delivery quantities, route statuses, and an interactive map view.
- **Design Choice: Real-Time Route Visibility.** The split-screen layout combines delivery information with map visualization, enabling drivers to track routes and delivery progress efficiently.

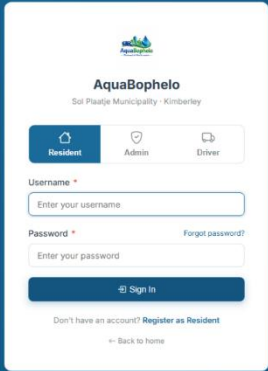
5.Delivery Status Screen



- **Description:** A delivery status interface allowing drivers to view and update scheduled deliveries, including notifications when no deliveries are assigned for the day.
- **Design Choice: Clear Status Communication.** The centered empty-state design immediately informs users about delivery availability while maintaining a clean and user-friendly interface.

DATA ENTRY FORMS

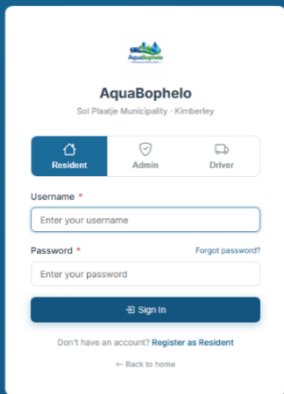
1. Registration Form



The registration form is displayed on a white card against a dark blue background. At the top, it features the AquaBophelo logo and the text 'Sol Plaatje Municipality - Kimberley'. Below this is a navigation bar with three tabs: 'Resident' (selected), 'Admin', and 'Driver'. The form fields include 'Username' with a red asterisk and a placeholder 'Enter your username', and 'Password' with a red asterisk, a placeholder 'Enter your password', and a 'Forgot password?' link. A blue 'Sign In' button is positioned below the password field. At the bottom, there is a link 'Don't have an account? Register as Resident' and a 'Back to home' link.

- **Description:** An onboarding screen capturing Name, Surname, Email, Password, and the crucial Area/Location.
- **Design Choice:** Error Prevention. By breaking registration into clear, sequential text fields and capturing "Area" immediately, this ensures that the system has the exact data needed to push targeted alerts later without overwhelming the new user.

2. Login Form



The login form is displayed on a white card against a dark blue background. It features the AquaBophelo logo and the text 'Sol Plaatje Municipality - Kimberley'. Below this is a navigation bar with three tabs: 'Resident' (selected), 'Admin', and 'Driver'. The form fields include 'Username' with a red asterisk and a placeholder 'Enter your username', and 'Password' with a red asterisk, a placeholder 'Enter your password', and a 'Forgot password?' link. A blue 'Sign In' button is positioned below the password field. At the bottom, there is a link 'Don't have an account? Register as Resident' and a 'Back to home' link.

- **Description:** A minimalist authentication screen with just Email, Password and a Login button.
- **Design Choice:** Recognition & Simplicity. It follows universal app standards. During a water shortage, users are likely stressed; a familiar, clutter-free login minimises cognitive load so they can access information instantly.

3. Resident Profile Update Form

The screenshot displays the AquaBophelo Resident Profile Update Form. The interface features a blue header with the AquaBophelo logo and 'Sol Plaatje Municipality' on the left, and user status 'HCI Resident' with 'RESIDENT' and 'Sign Out' buttons on the right. A sidebar on the left lists navigation options: Dashboard, Dam Levels, Track Trucks, Water Restrictions, Notifications, Submit Complaint, My Complaints, and My Profile (highlighted). The main content area is titled 'My Profile' and contains a profile card. The profile card shows a circular avatar with the letter 'H', the title 'HCI Resident', and the subtitle 'Resident - North Area'. Below this are input fields for First Name (HCI), Last Name (Resident), Email Address (hci_resident@example.com), Phone Number (0123456789), and a dropdown menu for Your Area (North Area). A 'Save Changes' button is located below the profile card. Below the profile card is a 'Change Password' section with input fields for Old password, New password, and New password confirmation, followed by an 'Update Password' button.

- **Description:** A screen allowing users to edit their existing contact and location details.
- **Design Choice:** User Control. Using explicit "Update" or "Save" buttons ensures users are fully aware when they are changing important contact information, preventing accidental data loss that could lead to missed SMS alerts.

4. Complaint Submission Form

The screenshot shows the 'Submit a Complaint' form in the AquaBophelo Resident Portal. The header includes the AquaBophelo logo and 'Sol Plaatje Municipality' on the left, and 'HCl Resident', 'RESIDENT', and 'Sign Out' buttons on the right. The left sidebar lists navigation options: Dashboard, Dam Levels, Track Trucks, Water Restrictions, Notifications, Submit Complaint (highlighted), My Complaints, and My Profile. The main content area is titled 'Submit a Complaint' with a subtitle 'Report water supply issues in your area. You will receive a reference number.' The form itself is a white box with the following fields: 'Complaint Type' (dropdown menu), 'Area' (dropdown menu), 'Incident Date' (text input with a date icon), and 'Description' (text area with a placeholder 'Please describe the issue in detail...'). At the bottom of the form are two buttons: 'Submit Complaint' and 'View My Complaints'.

- **Description:** An input interface for residents to report issues, featuring a category dropdown and a text description box.
- **Design Choice:** Efficiency. By providing categories (e.g., "Missed Delivery", "Pipe Leak") rather than just a blank box, you guide the user's input and this makes it faster for them to complain and easier for the Admin to sort and resolve.

5. Truck Registration Form (Admin)

The screenshot shows the 'Add Truck' form in the AquaBophelo Admin Portal. The header includes the AquaBophelo logo and 'Sol Plaatje Municipality' on the left, and 'Municipal Admin', 'Admin', and 'Sign Out' buttons on the right. The left sidebar lists navigation options: Dashboard, Dam Management, Truck Management (highlighted), All Trucks Map, Delivery Schedule, Create Driver, Driver Performance, User Management, Roles & Permissions, Account Status, Complaints, Restrictions, Announcements, Alert Management, SMS Log, Feedback, and Analytics. The main content area is titled 'Add Truck' with a subtitle 'Truck Management - Add Truck'. The form is a white box with the following fields: 'Truck id' (text input), 'Driver' (text input), 'Area' (dropdown menu with 'Central Area' selected), 'Route name' (text input), and 'Status' (dropdown menu with 'Active' selected). At the bottom of the form are two buttons: 'Save' and 'Cancel'.

- **Description:** An administrative screen to add a new water truck to the fleet, capturing the License Plate and assigning a Driver.
- **Design Choice: Constraint.** Using a dropdown menu to select the "Driver" instead of a text-typing field prevents spelling errors.

6. Truck Route Assignment Form (Admin)

Truck ID	Driver	Area	Route	Status	Actions
TRK-101	Bheki Themba Nkosi	NORTH	Water Depot → North Area (Galeshewe route)	ACTIVE	Edit Delete
TRK-102	Unsigned	CENTRAL	Water Depot → Central Area (CBD route)	ACTIVE	Edit Delete
TRK-103	Unsigned	SOUTH	Water Depot → South Area (Rondepan route)	ACTIVE	Edit Delete
TRK-104	Unsigned	WEST	Water Depot → West Area (Floors route)	ACTIVE	Edit Delete
TRK-105	Unsigned	NORTH	Water Depot → North Area (Greenpoint route)	ACTIVE	Edit Delete

- **Description:** An admin interface to schedule a specific truck to visit a specific area at a set time.
- **Design Choice: Logical Mapping.** By grouping inputs by Area, Truck, and Date/Time, the form matches the mental model of a logistics dispatcher, reducing the chance of double-booking a truck.

7. Water Level Update Form (Admin)

The screenshot displays the AquaBophelo Admin Portal. The header includes the logo, 'Municipal Admin', and a 'Sign Out' button. A sidebar on the left lists various administrative functions. The main content area is titled 'Add New Dam' and contains a form with the following fields: 'Name', 'Location', 'Capacity million litres' (with a value of 100,0), and 'Current level' (with a value of 50,0). The form includes 'Add Dam' and 'Cancel' buttons.

AquaBophelo
Gauteng Municipality

Municipal Admin **ADMIN** Sign Out

ADMIN PORTAL

- Dashboard
- Dam Management**
- Truck Management
- All Trucks Map
- Delivery Schedule
- Create Driver
- Driver Performance
- User Management
- Roles & Permissions
- Account Status
- Complaints
- Restrictions
- Announcements
- Alert Management
- SMS Log
- Feedback
- Analytics

Dam Management > Add Dam

Add New Dam

Name *

Location *

Capacity million litres *

100,0

Current level *

50,0

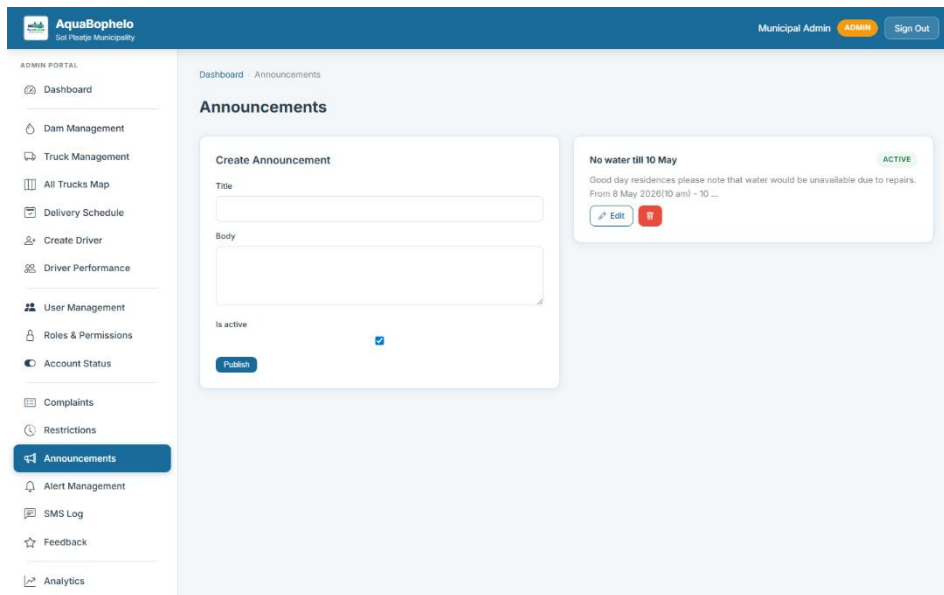
Add Dam Cancel

- **Description:** A secure form for admins to manually enter the current percentage of local dams.
- **Design Choice: Data Validation.** This form likely utilizes numeric-only fields to prevent typing errors (e.g., typing "O" instead of "0"), making sure that the data feeding into the predictive analytics dashboard is always accurate.

8. Historical Data Upload Form (Admin)

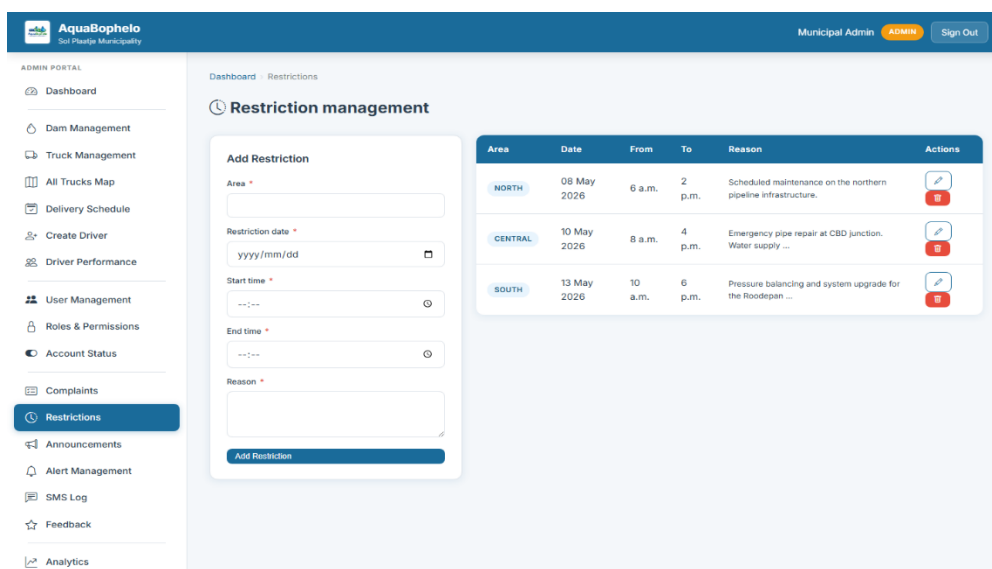
- **Description:** An interface allowing admins to upload past water statistics, likely via bulk file (CSV) or batch entry.
- **Design Choice: System Efficiency.** Allowing bulk uploads saves the admin from typing hundreds of past records manually, drastically reducing their workload and the potential for human error.

9. Notification Creation Form (Admin)



- Description: A communication tool where admins draft alerts, select the channel (SMS/Email) and pick the target Risk Zone.
- Design Choice: Targeted Precision. The ability to select specific "Zones" prevents alert fatigue. Residents only receive notifications relevant to their area, maintaining the system's credibility.

10. Water Restriction Schedule Form (Admin)



- Description: A scheduling tool to create upcoming water shedding events, detailing the start time, end time and affected area.

- **Design Choice: Standardization.** Utilizing calendar and clock widgets (date/time pickers) rather than free-text fields ensures that schedules are formatted perfectly when they are displayed to the residents on the frontend.

11. Driver Information Form

AquaBophelo
Sol Plaatje Municipality

Municipal Admin **ADMIN** Sign Out

ADMIN PORTAL

- Dashboard
- Dam Management
- Truck Management
- All Trucks Map
- Delivery Schedule
- Create Driver**
- Driver Performance
- User Management
- Roles & Permissions
- Account Status
- Complaints
- Restrictions
- Announcements
- Alert Management
- SMS Log
- Feedback
- Analytics

Dashboard Create driver account

Create driver account

Add a new driver and give them the username and password shown on the next screen. They log in at the same login page as everyone else.

Username *

Required. 150 characters or fewer. Letters, digits and @/./+/-/_ only.

First name

Last name

Email

Phone

Password *

* Your password must contain at least 8 characters.
* Your password can't be a commonly used password.

Password confirmation *

Enter the same password as before, for verification.

Truck

Assign a truck later (optional)

Create driver **Cancel**

Current drivers

- Sisekelo** — Sisekelo Preach Masombuka · sisekelopreachmasombuka@gmail.com
- bheki** — bheki Themba Nkosi · bheki@driver.com
- driver_test** — Driver Test · driver_test@example.com
- hcl_driver** — HCL Driver
- testdriver02** — Test Driver · testdriver02@example.com

- **Description:** A management screen used to capture and store driver-specific data, including contact details and professional records.
- **Design Choice: Accountability.** By centralizing driver data, the system ensures that every truck tracked on the live map is linked to a verified individual.

12. User Role Assignment Form

AquaBophelo

Botswana Municipality

Municipal Admin

Admin

Sign Out

ADMIN PORTAL

Dashboard

Dam Management

Truck Management

All Trucks Map

Delivery Schedule

Create Driver

Driver Performance

User Management

Roles & Permissions

Account Status

Complaints

Restrictions

Announcements

Alert Management

SMS Log

Feedback

Analytics

Admin

Users

Add User

Add New User

Create a new account for any role in the system

User Details

First Name *

Last Name *

e.g. Thabo

e.g. Mokoena

Username *

e.g. thabo.mokoena

Email Address

user@example.com

Role *

-- Select a role --

Password *

Min 6 characters

Confirm Password *

Repeat password

Create Account

Cancel

- Description: An administrative control panel where specific roles (Admin, Driver, Resident) are assigned to registered accounts.
- Design Choice: Security and Access Control. Using a clear selection interface ensures that sensitive actions like updating dam levels or re-routing trucks are restricted to authorized personnel, preventing unauthorized system changes.

13. Delivery Schedule Form

AquaBophelo

Botswana Municipality

Municipal Admin

Admin

Sign Out

ADMIN PORTAL

Dashboard

Dam Management

Truck Management

All Trucks Map

Delivery Schedule

Create Driver

Driver Performance

User Management

Roles & Permissions

Account Status

Complaints

Restrictions

Announcements

Alert Management

SMS Log

Feedback

Analytics

Dashboard

Delivery Schedule

Add Schedule

Truck

Area

From location

Water Depot

To location

Scheduled date

yyyy/mm/dd

Departure time

8:30 a.m.

Eta

10:30 a.m.

Distance km

0

Status

Scheduled

Add Schedule

Truck	Area	To	Date	Departs	ETA	Status
TRK-101	North	Galeshewe Community Hall	06 May	7 a.m.	8:30 a.m.	ON THE WAY
TRK-102	Central	CBD Water Point	06 May	8 a.m.	9 a.m.	SCHEDULED
TRK-103	South	Roodepan Reservoir	06 May	9 a.m.	10:30 a.m.	DELIVERED
TRK-104	West	Floors Community Centre	06 May	10 a.m.	11:30 a.m.	SCHEDULED
TRK-105	North	Greenpoint Tank	06 May	11 a.m.	noon	SCHEDULED
TRK-101	North	Galeshewe Community Hall	05 May	7 a.m.	8:30 a.m.	DELIVERED
TRK-102	Central	CBD Water Point	05 May	8 a.m.	9 a.m.	DELIVERED
TRK-103	South	Roodepan Reservoir	04 May	9 a.m.	10:30 a.m.	DELAYED

- Description: A scheduling interface used to input the specific dates, times, and areas for upcoming water truck deliveries.
- Design Choice: Visibility of System Status. The structured input format ensures that the data is standardized. This allows the system to populate the "Resident Delivery Schedule" screen accurately, reducing community anxiety through predictable service delivery.

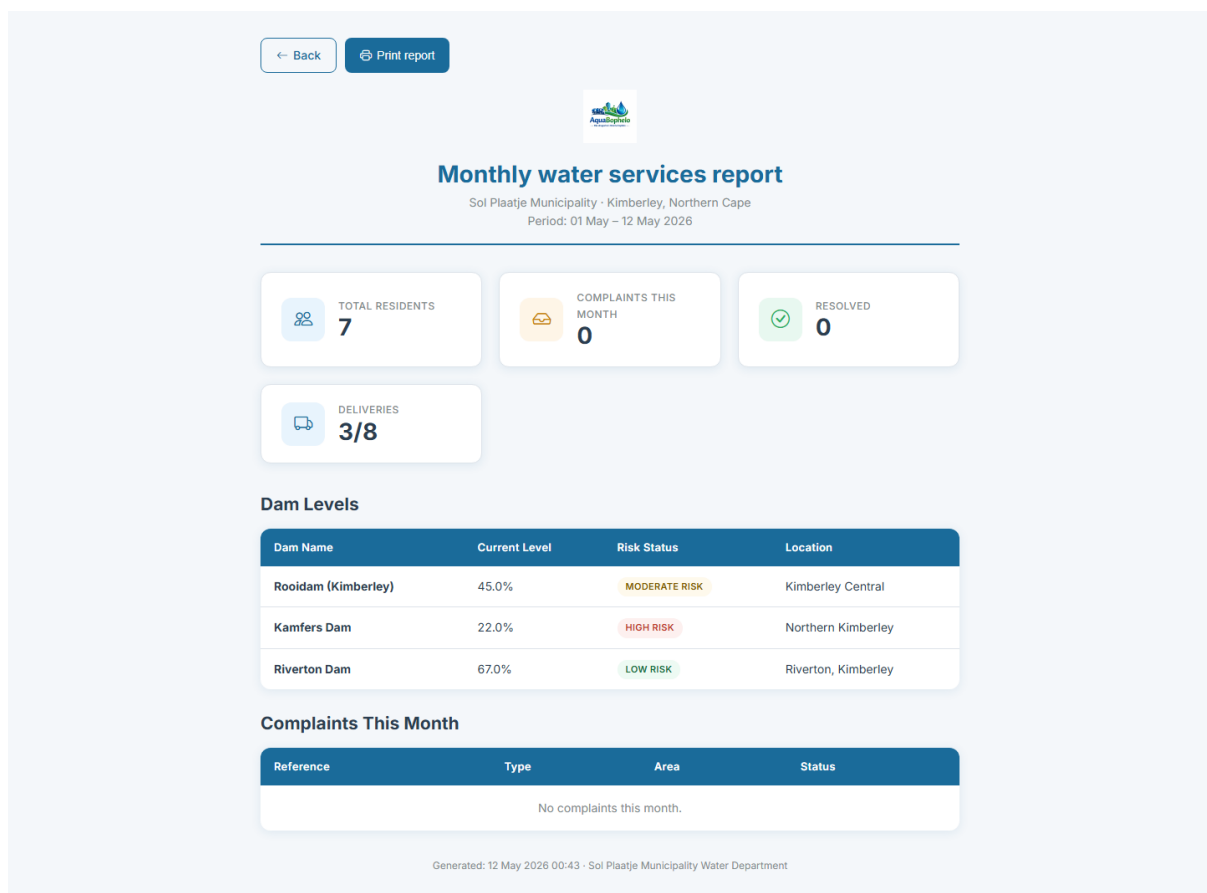
14. Feedback Form

The screenshot shows the 'Give Feedback' form within the AquaBophelo Resident Portal. The header includes the AquaBophelo logo and 'Sol Plange Municipality' on the left, and 'HCl Resident', 'RESIDENT', and 'Sign Out' buttons on the right. A left sidebar lists various portal features, with 'Give Feedback' highlighted. The main content area is titled 'Give Feedback' with a star icon and the subtitle 'Your feedback helps us improve water services for everyone'. The form itself is titled 'Share Your Experience' and contains the following fields: an 'Overall Rating' section with five star icons; a 'Category' dropdown menu with the placeholder '-- Select a category --'; a 'Comments' section with a text area and the prompt 'Tell us about your experience with water services in your area...'; and a 'Your Area' dropdown menu with 'North (Galesheve)' selected. A 'Submit Feedback' button is at the bottom of the form.

- Description: A simple communication interface for users to provide qualitative feedback on the system or water services.
- Design Choice: Iterative Improvement. This screen creates a direct loop between the community and the municipality. It allows admins to identify system pain points and improve the user experience over time.

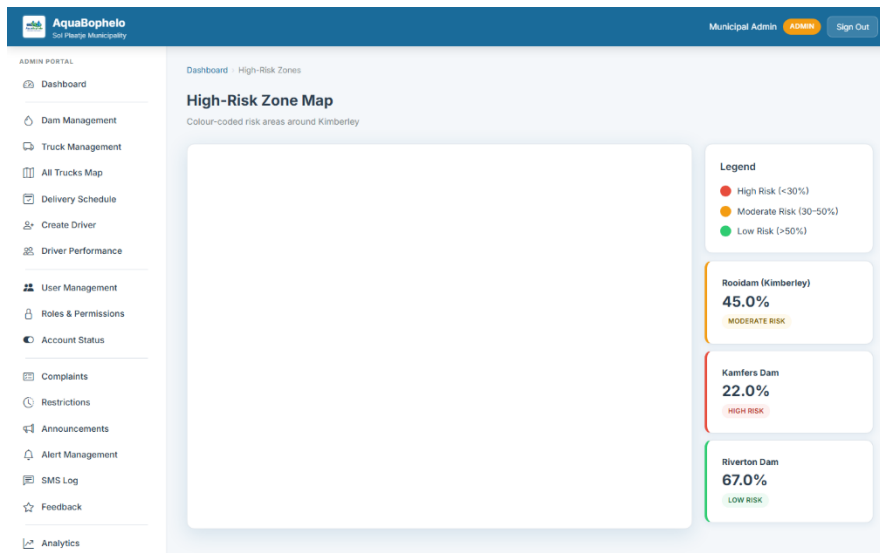
Report Generation Views

1.Monthly Water Services Report Screen



- **Description:** A report dashboard displaying monthly statistics related to residents, complaints, deliveries, and dam levels within Sol Plaatje Municipality.
- **Design Choice: Card-Based Information Display.** Summary cards were used to highlight important statistics at the top of the page, making information easier to scan quickly. Tables were included below for structured reporting and improved readability.

2.High-Risk Zone Map



- **Description:** A monitoring screen showing high-risk water shortage zones with colour-coded dam risk levels around Kimberley.
- **Design Choice: Colour-Coded Risk Visualization.** Red, orange, and green indicators were used to help administrators quickly identify low, moderate, and high-risk areas.

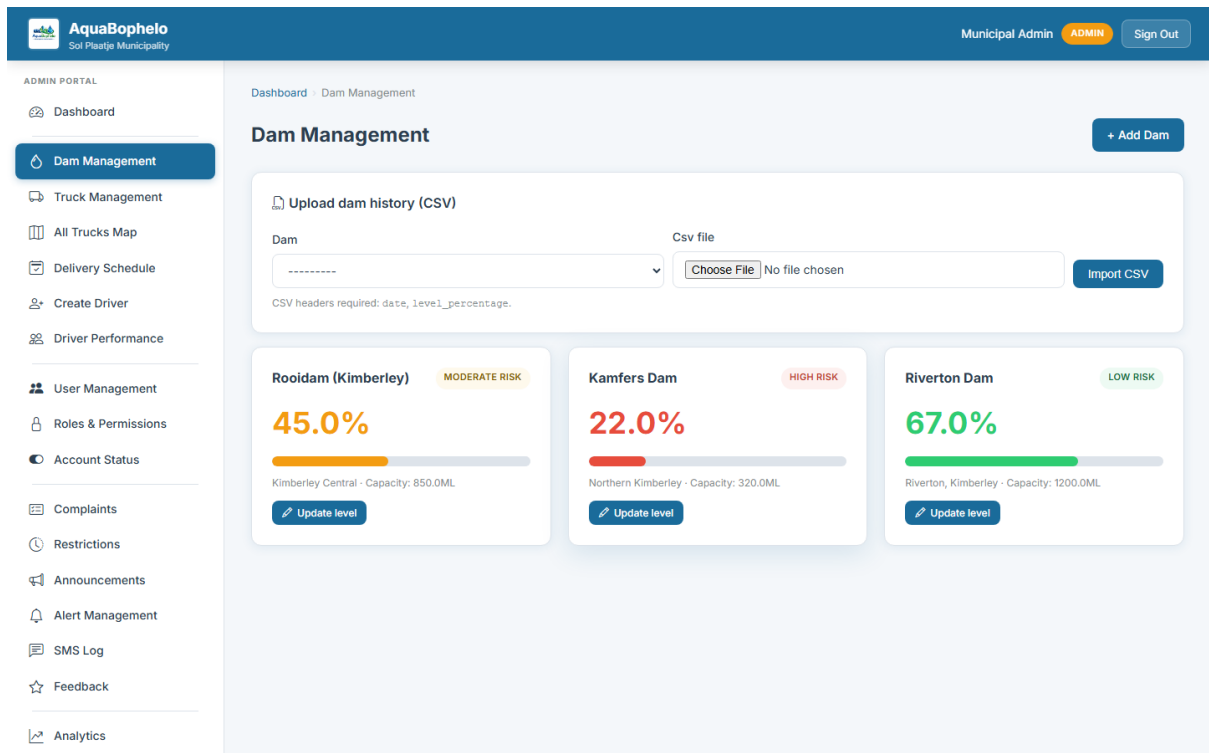
3.SMS Log Screen

The screenshot displays the 'SMS Log' dashboard for AquaBophelo. The left sidebar contains an 'ADMIN PORTAL' menu with options like Dashboard, Dam Management, Truck Management, All Trucks Map, Delivery Schedule, Create Driver, Driver Performance, User Management, Roles & Permissions, Account Status, Complaints, Restrictions, Announcements, Alert Management, SMS Log, Feedback, and Analytics. The main content area shows a table of simulated SMS messages sent to residents.

Recipient	Area	Message	Sent At
Resident One resident1@example.com	North	ALERT: Good day residences please note that water would be unavailable due to repairs. From 8 May 2026(10 am) - 10 May 2026 (8am)	07 May 2026 09:36
HCI Resident hci_resident@example.com	North	ALERT: Good day residences please note that water would be unavailable due to repairs. From 8 May 2026(10 am) - 10 May 2026 (8am)	07 May 2026 09:36
Resident Test resident_test@example.com	North	ALERT: Good day residences please note that water would be unavailable due to repairs. From 8 May 2026(10 am) - 10 May 2026 (8am)	07 May 2026 09:36
Cuba Cypher cubacypher999@gmail.com	North	ALERT: Good day residences please note that water would be unavailable due to repairs. From 8 May 2026(10 am) - 10 May 2026 (8am)	07 May 2026 09:36

- **Description:** A log management screen displaying SMS notifications sent to residents, including recipients, messages, and timestamps.
- **Design Choice: Structured Data Table Layout.** A table format was used to organise communication records clearly and improve tracking of sent notifications.

4. Dam Management Screen



- Description: A management dashboard allowing administrators to monitor dam levels, upload CSV files, and update water level information.
- Design Choice: Visual Water Level Cards. Individual cards with progress bars and risk labels were used to present dam conditions visually, improving monitoring efficiency and quick decision-making.

User Activity Report Screen


AquaBophelo
Sol Plaatje Municipality

Municipal Admin
ADMIN
Sign Out

ADMIN PORTAL

Dashboard

Dam Management

Truck Management

All Trucks Map

Delivery Schedule

Create Driver

Driver Performance

User Management

Roles & Permissions

Account Status

Complaints

Restrictions

Announcements

Alert Management

SMS Log

Feedback

Analytics

Admin Portal - User Activity Report

User Activity Report

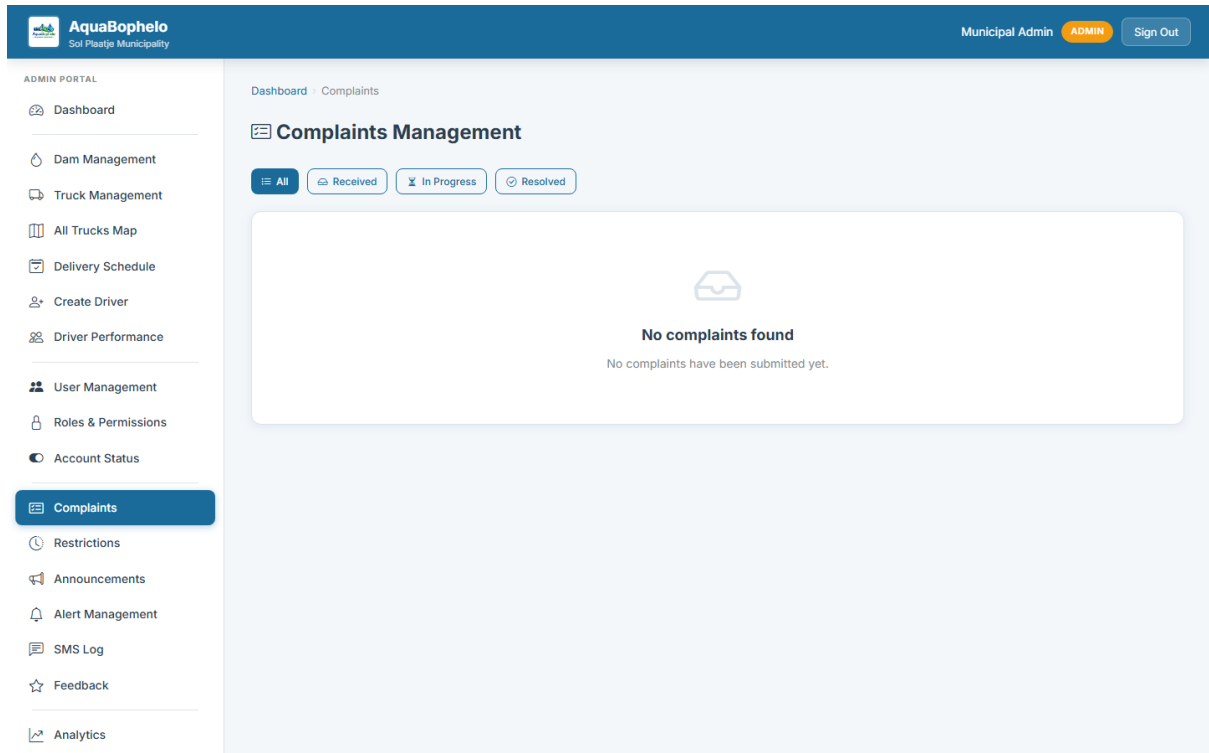
Recent system actions by all users

Time	User	Role	Action	Detail
12 May 00:38	thabo.m	RESIDENT	Login	Successful login
12 May 00:21	admin01	ADMIN	Update	Updated dam level for Gariep Dam
11 May 23:58	driver.b	DRIVER	Status Update	Delivery TRK-001 → Delivered
11 May 23:43	lindiwe.p	RESIDENT	Complaint	Submitted complaint WC-00019
11 May 22:43	admin01	ADMIN	Alert Sent	Critical alert sent to 142 residents
11 May 21:43	sipho.n	RESIDENT	Login	Successful login
11 May 20:43	driver.t	DRIVER	Location Update	GPS position updated
11 May 19:43	nomaa.d	RESIDENT	Logout	Session ended

Description: A reporting screen displaying recent user activities such as logins, updates, complaints, and alerts within the system.

- **Design Choice: Activity Tracking Table.** A structured table with role labels and activity details was used to improve system monitoring and accountability.

5.Complaints Management Screen



- **Description:** A complaints management page used to monitor and filter complaints submitted by residents.
- **Design Choice: Filter-Based Navigation.** Status filter buttons were added to allow administrators to sort complaints by category, improving usability and complaint tracking efficiency.

6.Driver Performance Screen


AquaBophelo
Sol Plaatje Municipality

Municipal Admin
ADMIN
Sign Out

ADMIN PORTAL

- Dashboard
- Dam Management
- Truck Management
- All Trucks Map
- Delivery Schedule
- Create Driver
- Driver Performance**
- User Management
- Roles & Permissions
- Account Status
- Complaints
- Restrictions
- Announcements
- Alert Management
- SMS Log
- Feedback
- Analytics

Dashboard > Driver Performance

Driver performance

Driver	Truck	Total Deliveries	Delivered	Delayed	Success Rate
Bheki Themba Nkosi bheki@driver.com	TRK-101	2	1	0	50%
Sisekelo Preach Masombuka sisekelopreachmasombuka@gmail.com	Not Assigned	0	0	0	N/A
Driver Test driver_test@example.com	Not Assigned	0	0	0	N/A
HCI Driver	Not Assigned	0	0	0	N/A
Test Driver testdriver02@example.com	Not Assigned	0	0	0	N/A

- **Description:** A performance tracking screen displaying driver delivery statistics, assigned trucks, delayed deliveries, and success rates.
- **Design Choice: Performance Monitoring Table.** A table layout with highlighted status indicators was used to allow administrators to compare driver performance quickly and effectively.

7.Delivery Schedule Screen


AquaBophelo
 Sol Plaatje Municipality

Municipal Admin
 ADMIN
Sign Out

ADMIN PORTAL

- Dashboard
- Dam Management
- Truck Management
- All Trucks Map
- Delivery Schedule**
- Create Driver
- Driver Performance
- User Management
- Roles & Permissions
- Account Status
- Complaints
- Restrictions
- Announcements
- Alert Management
- SMS Log
- Feedback
- Analytics

Dashboard / Delivery Schedule

Delivery Schedule

Add Schedule

Truck

Area

From location

To location

Scheduled date

Departure time

Eta

Distance km

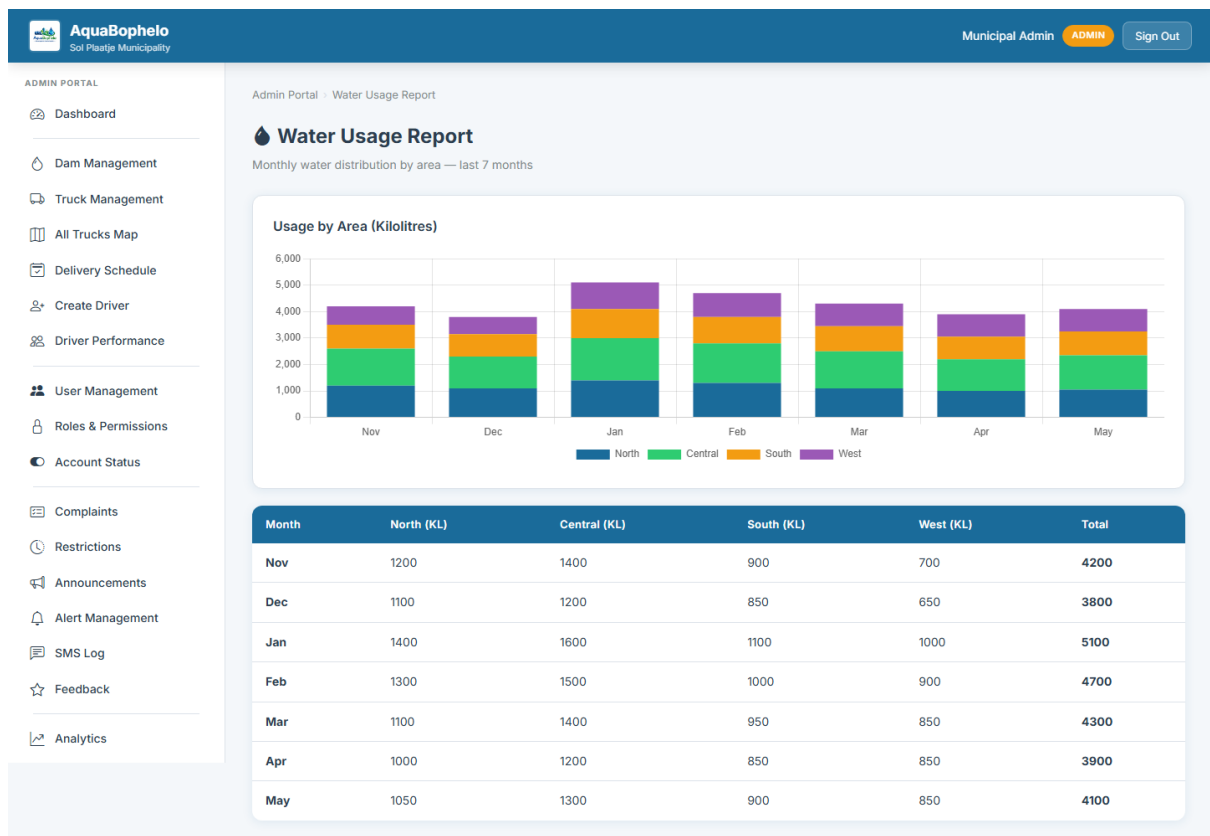
Status

Add Schedule

Truck	Area	To	Date	Departs	ETA	Status
TRK-101	North	Galeshewe Community Hall	06 May	7 a.m.	8:30 a.m.	ON THE WAY
TRK-102	Central	CBD Water Point	06 May	8 a.m.	9 a.m.	SCHEDULED
TRK-103	South	Roodepan Reservoir	06 May	9 a.m.	10:30 a.m.	DELIVERED
TRK-104	West	Floors Community Centre	06 May	10 a.m.	11:30 a.m.	SCHEDULED
TRK-105	North	Greenpoint Tank	06 May	11 a.m.	noon	SCHEDULED
TRK-101	North	Galeshewe Community Hall	05 May	7 a.m.	8:30 a.m.	DELIVERED
TRK-102	Central	CBD Water Point	05 May	8 a.m.	9 a.m.	DELIVERED
TRK-103	South	Roodepan Reservoir	04 May	9 a.m.	10:30 a.m.	DELAYED

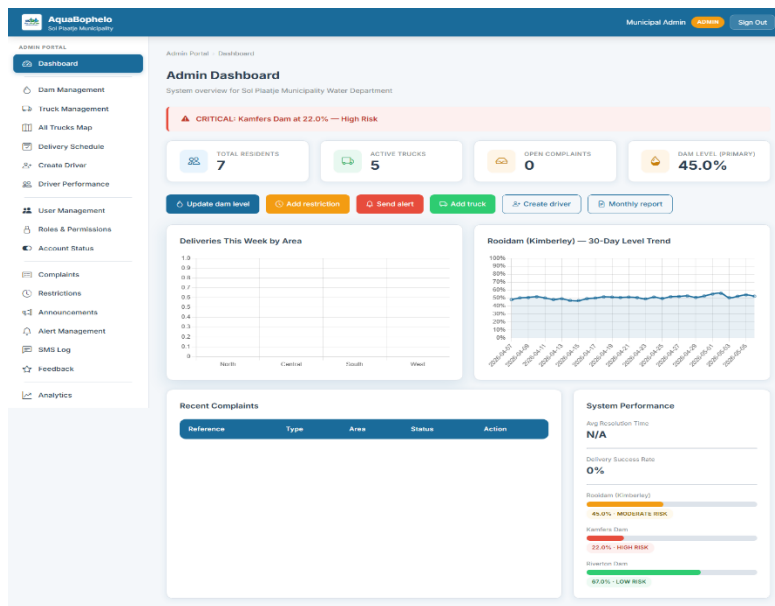
- **Description:** A scheduling interface allowing administrators to assign trucks, delivery locations, dates, and delivery times.
- **Design Choice: Form-and-Table Combination Layout.** The scheduling form was positioned alongside the delivery table to allow users to add and review schedules

Water Usage Report Screen



- **Description:** An analytics dashboard displaying monthly water distribution statistics across different municipal areas.
- **Design Choice: Data Visualisation with Charts.** Bar charts and tables were combined to improve understanding of water usage trends and support better analysis of distribution patterns.

Admin Dashboard

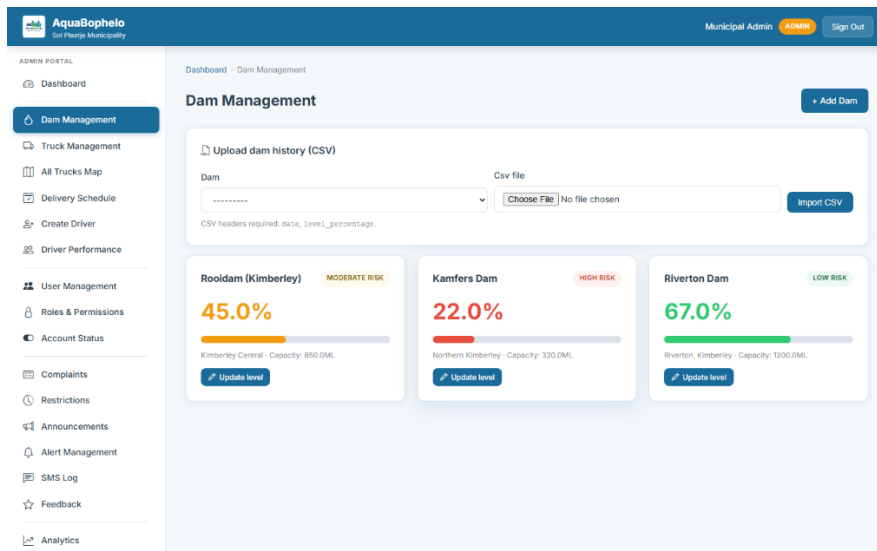


Description

This screen provides the municipal administrator with a complete overview of the water delivery system. It displays key system statistics such as total residents, active trucks, open complaints, and dam level percentage. The dashboard also shows alerts for critical dam levels, weekly delivery charts, dam trend graphs and recent complaints. The purpose of this screen is to support quick decision-making and monitoring of water shortages.

Design Choice: Consistency and Standards. The sidebar navigation remains visible across all admin screens, helping admins maintain familiarity while navigating the system.

Water Data Management Panel



Description

This screen allows managers to monitor dam water levels and risk conditions across different locations. Each dam is displayed inside a card showing the current percentage level, storage capacity, and risk category (Low, Moderate, or High Risk). The screen also allows administrators to upload dam history using CSV files and update dam levels manually.

Design Choice (Visibility of System Status): The use of Progress Bars and Colour-Coding (Orange for Moderate, Red for High, Green for Low) gives the admin instant feedback. They can see which dams are in danger just by looking at the colours without reading the numbers.

Truck Management

The screenshot shows the AquaBophelo Truck Management dashboard. The sidebar on the left contains navigation links for Dashboard, Dam Management, Truck Management, All Trucks Map, Delivery Schedule, Create Driver, Driver Performance, User Management, Roles & Permissions, Account Status, Complaints, Restrictions, Announcements, Alert Management, SMS Log, Feedback, and Analytics. The main content area is titled 'Truck Management' and features a 'View map' button and a '+ Add Truck' button. Below this is a table listing trucks with columns for Truck ID, Driver, Area, Route, Status, and Actions. The table shows five trucks: TRK-101, TRK-102, TRK-103, TRK-104, and TRK-105, all with 'ACTIVE' status and 'EGR' actions.

Truck ID	Driver	Area	Route	Status	Actions
TRK-101	Bheki Themba Nikosi	NORTH	Water Depot → North Area (Galesheve route)	ACTIVE	EGR EGR
TRK-102	Unassigned	CENTRAL	Water Depot → Central Area (CBD route)	ACTIVE	EGR EGR
TRK-103	Unassigned	SOUTH	Water Depot → South Area (Roodepan route)	ACTIVE	EGR EGR
TRK-104	Unassigned	WEST	Water Depot → West Area (Floors route)	ACTIVE	EGR EGR
TRK-105	Unassigned	NORTH	Water Depot → North Area (Greenpoint route)	ACTIVE	EGR EGR

Description

This screen allows the admin to monitor the water delivery fleet and assign drivers to specific municipal routes. Each truck is displayed in a table showing the Truck ID, the assigned driver's name, the geographic area, and the specific delivery path. The screen also allows administrators to add new trucks to the system, edit existing route details, or delete inactive vehicles from the list.

Design Choice (Recognition over Recall): The use of a table with clearly labelled headers ensures that admins don't have to memorise which truck belongs to which area. All data is visible at once.

User Management Panel

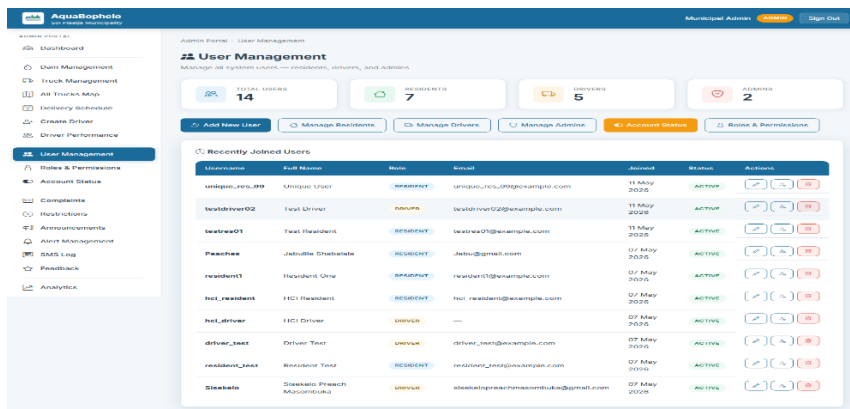
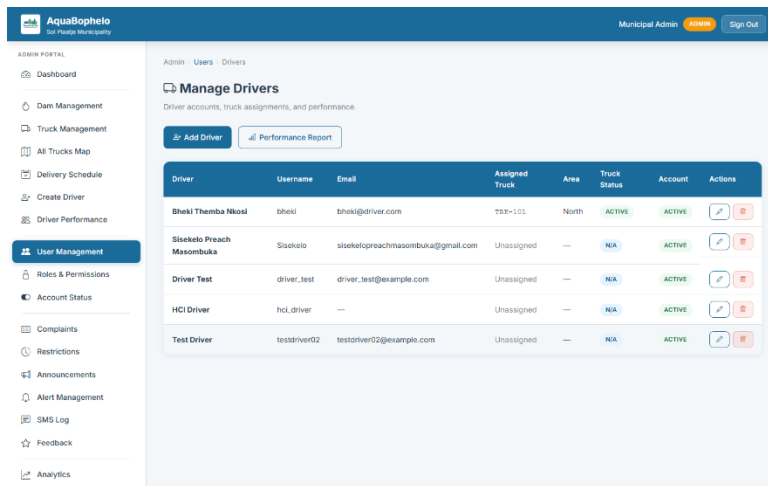
Description

This screen allows administrators to manage and track every person registered in the system, including residents, drivers, and other admins. The panel displays high-level statistics cards

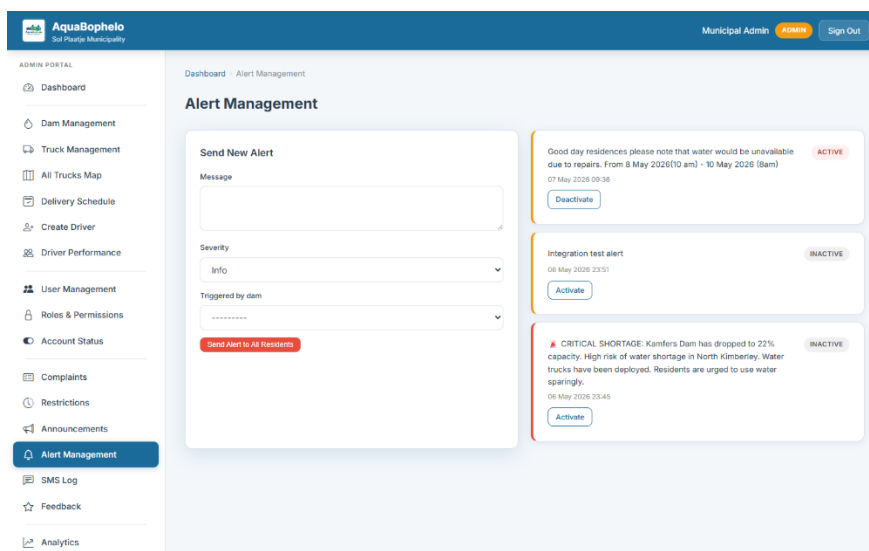
showing the total number of users and specific counts for each role to provide a quick overview of the system's population.

Design Choice (Visibility of System Status): The Summary Cards at the top provide immediate feedback on the "health" of the system (e.g., how many drivers are available) without the admin having to filter or count manually.

The screenshot displays the AquaBophele Admin Portal interface. The top navigation bar includes the logo, 'Municipal Admin', and 'Sign Out' buttons. A sidebar on the left lists various administrative functions such as Dashboard, User Management, Roles & Permissions, and Analytics. The main content area is titled 'Add New User' and contains a form for creating a new account. The form includes fields for First Name, Last Name, Username, Email Address, Role (a dropdown menu), Password, and Confirm Password. A 'Create Account' button is located at the bottom of the form.



Alert Management Screen



Description

This screen allows managers to create and broadcast emergency notifications to residents regarding water shortages or repairs. Administrators can type a custom message, select the severity level (like Info or Danger), and link the alert to a specific

dam. The screen also keeps a history of sent alerts, allowing managers to see which are currently "Active" and giving them the option to deactivate or reactivate them as needed.

Design Choice (User Control and Freedom): The inclusion of "Activate" and "Deactivate" buttons on every alert card allows the manager to quickly stop a notification if a repair finishes early, giving them full control over the communication flow.

Audit Logs Screen

 AquaBophelo
Sol Plaatje Municipality

Municipal AdminADMINSign Out

ADMIN PORTAL

Dashboard

Dam Management

Truck Management

All Trucks Map

Delivery Schedule

Create Driver

Driver Performance

User Management

Roles & Permissions

Account Status

Complaints

Restrictions

Announcements

Alert Management

SMS Log

Feedback

Analytics

Admin Portal > Audit Logs

 Audit Logs

System activity trail — all significant actions recorded

Time	User	Action	Module	Detail	Severity
12 May 2026 00:42	admin01	UPDATE	Dam	Gariep Dam level set to 31%	WARNING
12 May 2026 00:29	admin01	CREATE	Alert	Critical alert sent to 142 residents	DANGER
12 May 2026 00:14	driver.b	UPDATE	Delivery	Delivery status → Delivered (TRK-001)	SUCCESS
11 May 2026 23:44	admin01	CREATE	Truck	New truck TRK-005 registered	INFO
11 May 2026 22:44	admin01	DELETE	Restriction	Restriction #12 removed	WARNING
11 May 2026 21:44	system	LOGIN	Auth	Admin login from 196.22.x.x	INFO
11 May 2026 19:44	admin01	UPDATE	User	Driver account driver.t role changed	WARNING
11 May 2026 16:44	system	BACKUP	System	Automatic database backup completed	SUCCESS

Description

This screen provides a detailed history of every significant action taken within the system, acting as a digital paper trail. It records the time of the action, which user performed it (e.g., admin01), the type of action (CREATE, UPDATE, DELETE), and specific details about what was changed.

Design Choice (Recognition over Recall): The Severity column uses colour-coded labels (Success, Warning, Danger). Instead of reading every single detail, an admin can quickly scan for "Danger" or "Warning" tags to find potential issues or unauthorised changes.

Analytics Screen



Description: This screen provides high-level insights and trend analysis for the entire municipality. It uses visual charts to show water usage trends, complaint categories (via a donut chart), delivery success rates by area, and dam level declines over time

Design Choice (Aesthetic and Minimalist Design): The use of Data Visualisation (Graphs and Charts) instead of raw data tables helps the admin process complex information quickly. It highlights the most important trends without cluttering the screen with numbers.

USER FEEDBACK

User Integration

1. Registration Process Length

User Feedback: Naledi commented: *“There are too many fields on one screen, and I was not sure which ones I had to complete first.”* She felt that the registration form required a step-by-step process.

Design Iteration: Introduction of her multi-step registration process in one form, but with sections such as personal information, contact details and security setup

Why: This will reduce cognitive overload and make the process of registration faster and easier to complete.

3. Driver Dashboard Lacked Priority Information

- User Feedback: Kabelo mentioned, *“When I log in, I want to immediately know if I have a truck assigned instead of reading through the page.”* Drivers preferred important delivery information to stand out more clearly.
- Design Iteration: A large central notification card was added to the dashboard to display truck assignment status prominently.
- Why: This improves Efficiency and Visual Hierarchy by presenting the driver’s most important task information first.

4. GPS Location Submission Needed faster Interaction

- User Feedback: Sipho explained, *“While driving, it is difficult to focus on small buttons. I need the location update action to be quick and obvious.”* He felt that the process regarding the update needs to require less effort while on the move.
- Design Iteration: The Submit Location button was made bigger and placed directly under the coordinate fields, while the “Use My GPS Location” button was made more prominent.
- Why: This improves Accessibility and Ease of Interaction by reducing the time needed to complete updates during deliveries.

5. The "Dashboard" vs. "Home" Redundancy

- **User Feedback: Lerato** noted, *“When the water is off, I am in a hurry. I don't want to see a 'Welcome' message; I just want to know if the truck is coming.”* Users feel confused by having both a Home Page and a Dashboard in high-stress situations.

- **Design Iteration:** Merge the Home Page and Dashboard. When a resident logs in, the very first thing they see is the "Water Status" of their specific area.
- **Why:** This improves **Efficiency**. It removes one click from the user's journey to find critical information.

6. Highlighting "Time to Arrival"

- **User Feedback:** Lerato specifically stated, *"The map is nice, but the arrival time is too small. I have to squint to see it while I'm getting my buckets ready."*
- **Design Iteration:** On the Water Truck Tracking Screen, use a large, high-contrast countdown timer (e.g., **"Arrival in 12 Mins"**) in a bold color like orange or blue.
- **Why:** Highlighting the ETA reduces user anxiety and follows the principle of **Visual Hierarchy**, making the most vital data the most prominent.

3. Progressive Disclosure on Forms

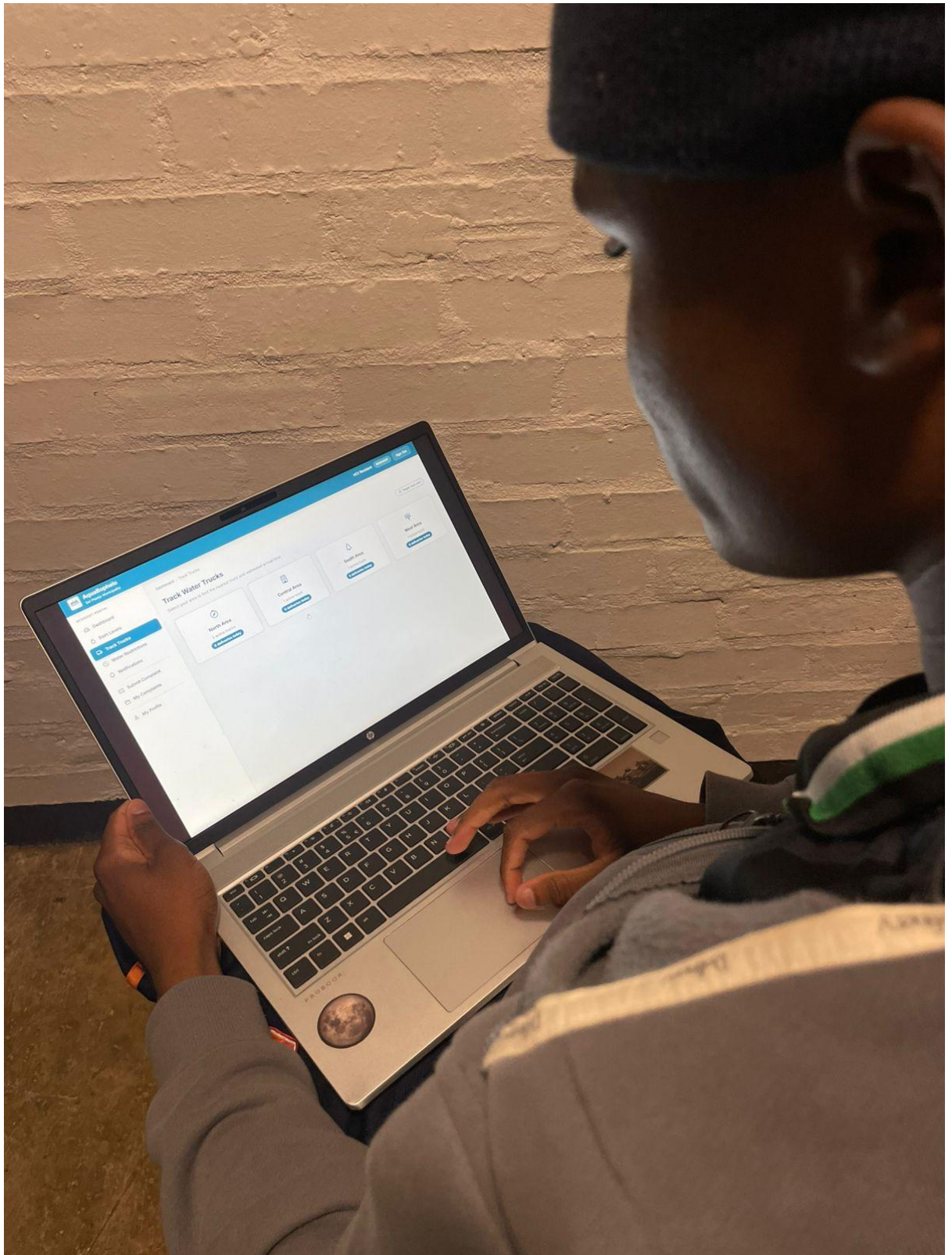
- **User Feedback:** Sbani commented on the registration process, saying, *"There are so many boxes to fill in all at once, it feels like a lot of work just to sign up."* Mr. Jansen also noted that the Route Assignment form looked "cluttered."
- **Design Iteration:** Use **Progressive Disclosure** for complex forms. Break Registration or Route Assignment into 2 or 3 steps (e.g., Step 1: Personal Info, Step 2: Location).
- **Why:** This manages **Cognitive Load**. Showing only a few fields at a time prevents users from feeling overwhelmed and reduces input errors.

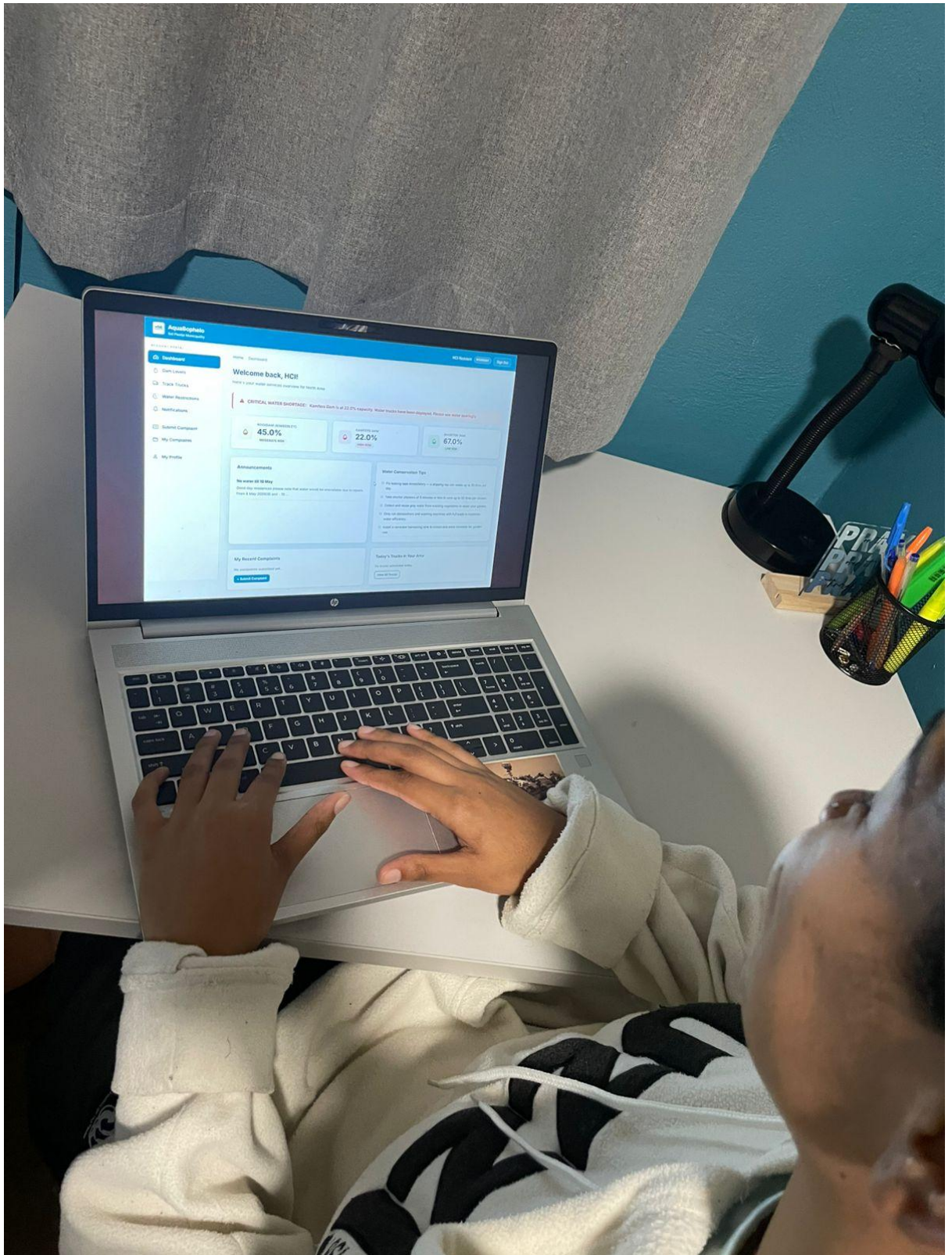
7. Color-Coded Status Indicators

- **User Feedback:** Mr. Jansen mentioned, *"I have to read every single line to see which dam is low. I wish I could just see the danger spots immediately."*
- **Design Iteration:** Implement a **"Traffic Light" system**:
 - **Green:** Water levels > 70% / Truck on time.
 - **Yellow:** Water levels 30-70% / Delivery delayed.
 - **Red:** Water levels < 30% / Shortage alert.
- **Why:** This supports **Recognition over Recall**. Users instantly understand the severity through color before reading the text.

8. Accessibility for Diverse Literacy Levels

- **User Feedback:** During testing, **Sbani** suggested, *"Some of my neighbours might struggle with all this English text. Can we make it easier for people who just want to see what to do?"*
- **Design Iteration:** Incorporate more **Icons and Visual Aids**. Use illustrations (e.g., a bucket icon vs. a hose icon) for tips and specific icons for complaint types (e.g., a leaking tap icon).
- **Why:** This improves **Inclusivity** and accessibility, ensuring the system is usable regardless of reading speed or primary language.



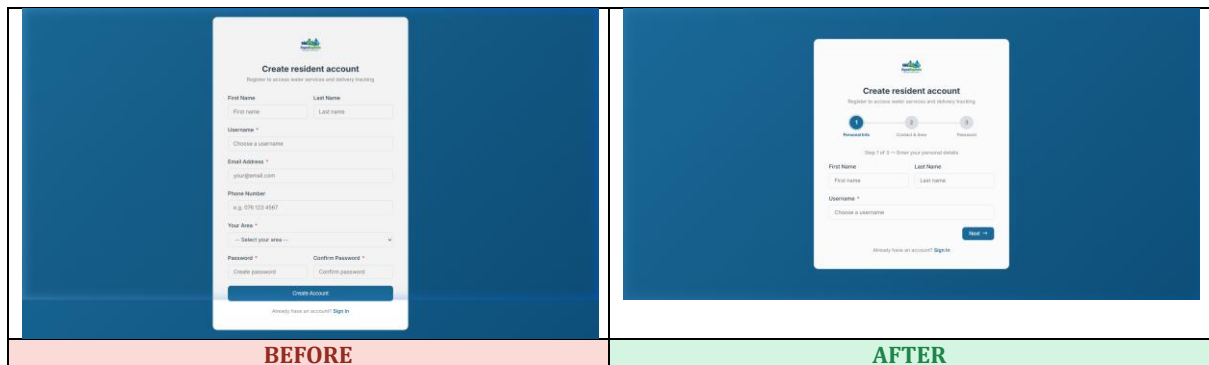


Design iteration

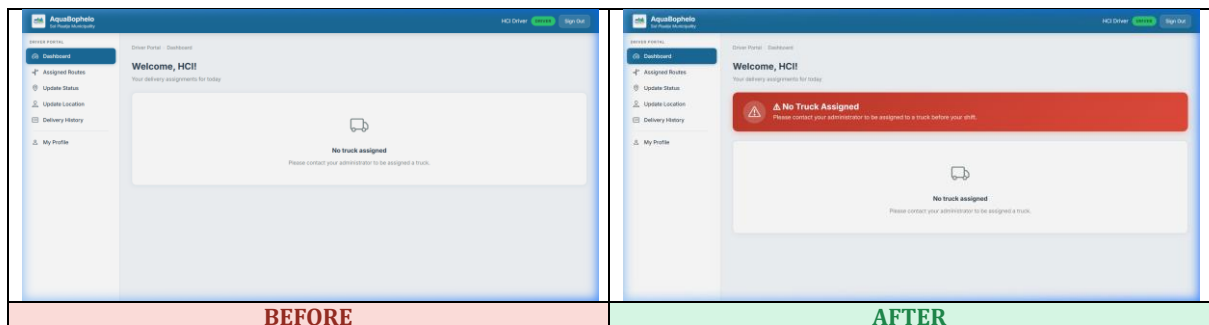
Summary Changes

	Feature	Change Made
1	Registration Form	3-step form + green submit button top-right
2	Driver Dashboard	Large colour-coded truck status banner
3	Driver Location Update	Bigger GPS buttons + success message
4	Resident Dashboard	Water status moved to very top
5	Truck Tracking	Bold 56px orange ETA countdown
6	Dam Management (Admin+Res)	Traffic light ● ● ● colour coding + legend
7	Add New User	Red 'Clear Form' button with flash confirms
8	Manage Residents	Checkbox bulk-delete toolbar

Before and After

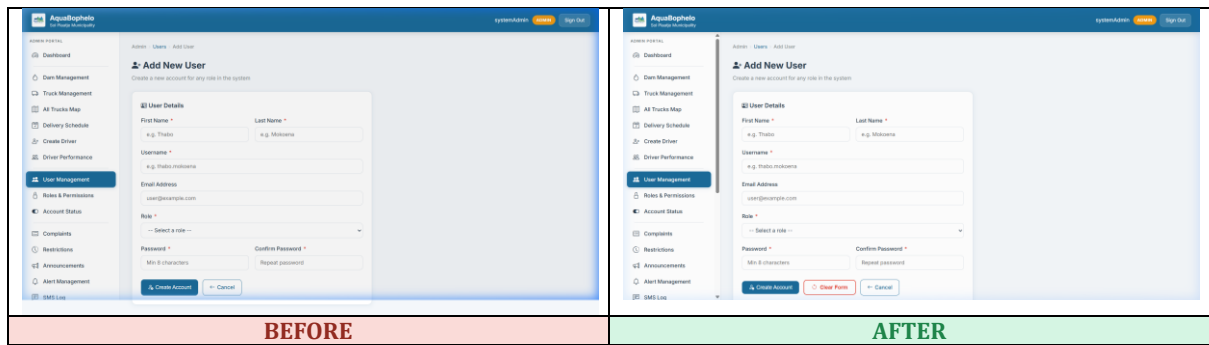


Before and After

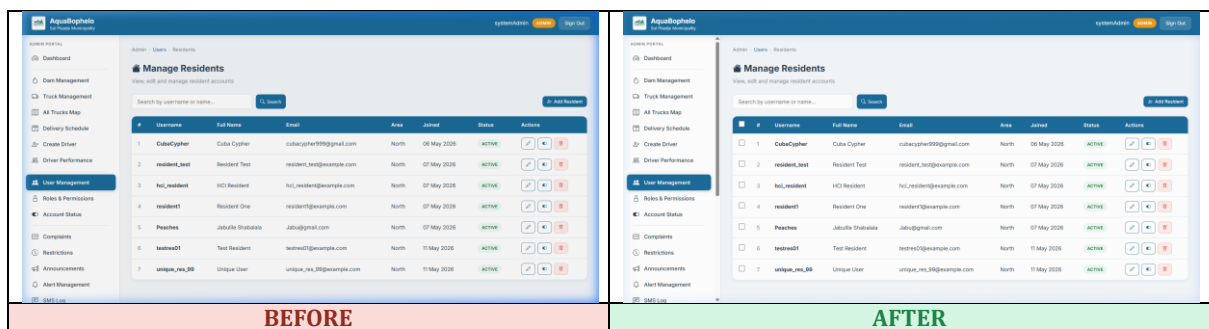


The figure compares two versions of the Aquaphoto user interface. The left side, labeled 'BEFORE', shows a dashboard with a sidebar on the left containing links like 'Dashboard', 'Data Levels', 'Track Trucks', 'Water Restrictions', 'Notifications', 'Submit Complaint', and 'My Complaints'. The main content area has a 'Welcome back, HCII!' message, a 'CRITICAL WATER SHORTAGE' alert, and three water status cards. The right side, labeled 'AFTER', shows a more modern dashboard with a top navigation bar and a main content area. It also features a 'Welcome back, HCII!' message, a 'CRITICAL WATER SHORTAGE' alert, and three water status cards. The 'AFTER' version has a cleaner, more organized layout with a focus on the critical water shortage alert.

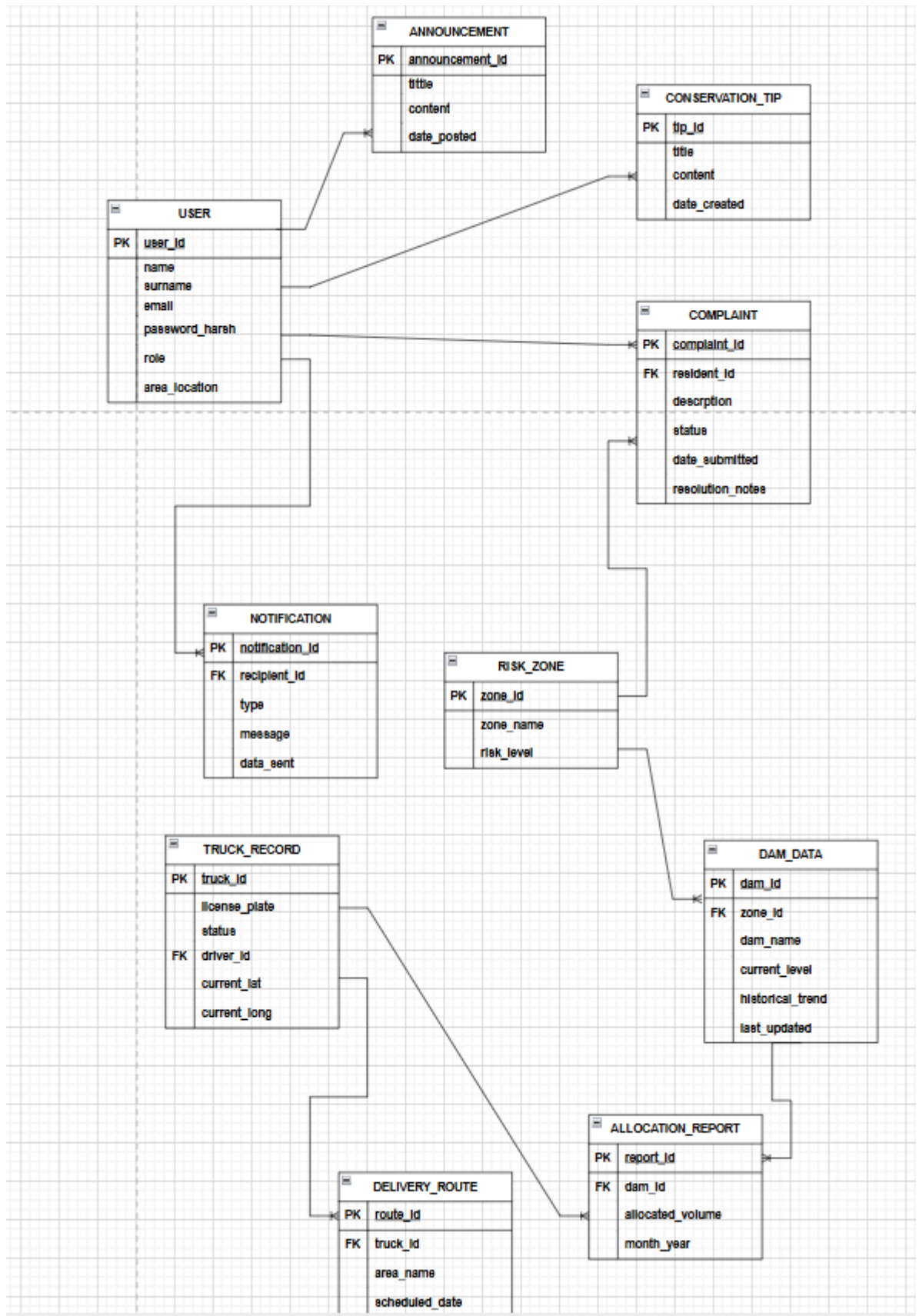
Before and After



Before and After



DATABASE DESIGN



RELATIONSHIP JUSTIFICATION

1. USER ↔ COMPLAINT Relationship

- One **USER** can submit many **COMPLAINTS**
- One complaint belongs to one resident.

Residents must be able to:

- submit complaints,
- track complaint status,

-The **foreign key resident_id** in the **COMPLAINT entity** ensures that each complaint is connected to the right resident resident accountability and traceability.

This relationship is required because it:

- Enables admins to manage complaints per resident
- Supports complaint history tracking
- Improves communication between residents and municipality

2. USER ↔ NOTIFICATION Relationship

- One **USER** can receive many **NOTIFICATIONS**

The system must:

- send SMS notifications
- send email notifications
- generate water shortage alerts.

-The **recipient_id foreign key** ensures notifications are sent to the right resident.

This relationship is required because it:

- Supports real-time communication
- Reduces uncertainty during shortages
- Improves user awareness and efficiency

-Announcements are connected to users because admins are responsible for posting and managing official communication.

3. USER ↔ ANNOUNCEMENT Relationship

- Admin users create announcements that are viewed by residents.

This relationship supports the requirement:

- “The system shall allow residents to view important announcements.”

Municipal administrators need a way to communicate public information such as:

- maintenance notices
- delivery updates
- emergency alerts.

This relationship is required because it:

- Centralizes public communication
- Improves transparency
- Supports information sharing

4. USER ↔ CONSERVATION_TIP Relationship

- Users access multiple conservation tips.

This relationship supports the requirement:

- “The system shall provide water conservation tips.”

This relationship is required because it:

- Encourages responsible water usage
- Supports awareness and education
- Helps users respond to water shortages effectively

5. TRUCK_RECORD ↔ DELIVERY_ROUTE Relationship

- One truck can have many delivery routes.

Admins must:

- allocate delivery routes
- schedule deliveries
- track truck progress.

-The **truck_id foreign key** connects each route to the right truck.

This relationship is required because it:

- Enables route scheduling
- Supports real-time delivery tracking
- Allows admins to monitor truck operations

6. USER ↔ TRUCK_RECORD Relationship

- One driver (USER role = Driver) can be assigned to one or multiple trucks.

The system supports multiple user roles:

- Resident
- Admin
- Driver.

-Drivers are responsible for operating water trucks and updating delivery progress.

This relationship is required because it:

- Supports driver assignment
- Tracks driver performance
- Connects truck operations to responsible personnel

-It enhances accountability and supports delivery monitoring.

7. RISK_ZONE ↔ DAM_DATA Relationship

- One risk zone can contain multiple dam records.

This relationship supports:

- water shortage risk analysis,
- high-risk zone monitoring,
- water monitoring analytics.

This relationship is required because it:

- Helps identify high-risk communities
- Supports predictive analysis
- Enables better municipal planning

8. DAM_DATA ↔ ALLOCATION_REPORT Relationship

- One dam can generate many allocation reports.

This relationship supports the requirement:

- “The system shall create monthly water allocation reports.”
-Allocation reports depend on current and historical dam levels.

This relationship is required because it:

- Supports water distribution planning
- Tracks water usage over time
- Assists decision-making for municipalities

Technology Choice program

Frontend: React.js + Tailwind CSS

React is the pick here because your system has multiple dynamic views a live dam level graph, a real-time truck map, a complaint tracker all updating without full page reloads. React's component model lets your team split the Resident Portal and Admin Dashboard into reusable pieces, which also makes collaboration cleaner. Tailwind handles responsiveness natively, which is critical since your non-functional requirements explicitly demand mobile support. Rural residents in Ritchie are more likely on a phone than a laptop, so a mobile-first UI isn't optional it's the whole point.

Backend: Node.js + Express

Node.js is a strong fit because it's JavaScript, which means your team only needs to master one language across frontend and backend. Express keeps the API structure lightweight and flexible you're not building a massive enterprise app, you're building a focused municipal tool, and Express gives you exactly the structure you need without bloat. It handles your REST endpoints cleanly: /api/dam-levels, /api/trucks, /api/complaints, and so on. For authentication, **JWT (JSON Web Tokens)** is the standard modern approach for role-based access. When a resident logs in they get a token tagged RESIDENT, an admin gets ADMIN, a driver gets DRIVER and your backend checks that token before serving any sensitive route. This maps perfectly to your functional requirement of role-based permissions.

Database: PostgreSQL + Prisma ORM

PostgreSQL is relational, which fits your data model perfectly users, trucks, complaints, dam readings, and delivery routes all have clear relationships between them. It's also the industry standard, open-source, and runs well on free-tier hosting like Render or Railway. Prisma ORM sits on top and gives you type-safe queries in JavaScript less raw SQL to debug, and it generates your database schema from a clean schema file which is easier to manage as a team.

Mapping Leaflet.js

Leaflet is open-source, lightweight, and specifically designed for interactive maps in the browser. It supports custom markers (perfect for truck icons), real-time position updates via JavaScript, and works beautifully on mobile. Since you're doing GPS simulation rather than actual live GPS hardware, Leaflet lets you push coordinates from the backend on a timer and watch the truck marker move on the map clean simulation of real-world tracking.

Data Visualisation: Chart.js

Chart.js handles your dam level trend graphs line charts over time, percentage gauges, historical comparisons. It's the easiest charting library to integrate with React and has excellent mobile rendering. Your admin analytics dashboard will lean on this heavily.

Deployment: Vercel (Frontend) + Render (Backend) + Supabase (PostgreSQL)

All three have generous free tiers, which matters for a student project. Vercel deploys React apps in minutes. Render hosts your Node.js API. Supabase gives you a managed PostgreSQL instance with a visual dashboard to inspect your data extremely useful during development. This whole stack runs without spending a cent, which makes it realistic to actually ship.

